

Kabul University
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Design and Implementation of a Medical Exit Exam Preparation Mobile Application

A Project Submitted in Partial Fulfillment
of the Requirements for the Degree of
Bachelor of Science in Software Engineering

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بسم الله الرحمن الرحيم

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APPROVAL PAGE

This project report titled “Design and Implementation of a Medical Exit Exam Preparation Mobile Application” is hereby approved as a satisfactory report in partial fulfillment of the requirements for the Bachelor of Science in Software Engineering.

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We, Aref Sediqi, P26000174; Azizullah Yosufi, P07003757; and Najeebullah Karimi, P01056335, declare that this final year project report titled "Design and Implementation of a Medical Exit Exam Preparation Mobile Application" is our original work. All sources, references, and quoted materials have been duly acknowledged and cited according to APA standards.

This work has not been previously submitted, in whole or in part, for any other degree or diploma at Kabul University or any other institution.

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ABSTRACT

Write 200–300 words here: Problem, Methodology, Results, Conclusion

Keywords: [Keyword1, Keyword2, Keyword3, Keyword4, Keyword5]

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CHAPTER 1: INTRODUCTION

This chapter introduces the project and the context in which it was developed. It begins by explaining the Afghan Medical Exit Exam and the preparation challenges faced by its candidates, then presents the problem statement, objectives, corresponding research questions, scope, limitations, and significance of the project. It also summarizes the research and development methodologies, states the project outcomes, and outlines the organization of the thesis. Together, these sections establish the purpose, direction, boundaries, and expected contribution of an offline-capable and exam-oriented mobile preparation application.

1.1 Background

In Afghanistan, medical graduates do not move from graduation into professional practice through the university degree alone. They must also pass a national Exit Exam before they can obtain a medical practice permit and register as practitioners. According to Rahimi et al. (2026), the current official name of this exam is the National Medical Licensing Examination (NMLE), although it is still widely referred to in public usage as the Exit Exam. It functions as a licensing requirement for curative medicine and stomatology graduates who intend to work in the country, including graduates from local and foreign institutions. The same review explained that the exam was initially known as the Exit Exam, was later renamed the Afghanistan Medical Licensing Examination (AMLE), and then became known as the NMLE. The exam has been reported as a broad multiple-choice assessment with 160 questions, 360 total marks, a passing score of 216 (60%), and a four-hour time limit. Its content is divided between paraclinical subjects, which make up 30% of the exam, and clinical subjects, which make up 70%. The paraclinical part covers the same broad subjects for both curative medicine and stomatology, including anatomy, physiology, pathology, pharmacology, microbiology, and medical ethics, although the exact content can differ according to the medical or dental focus of the exam. For curative medicine, the clinical part includes general internal medicine, surgery, obstetrics and gynecology, pediatrics, ENT (ear, nose, and throat), public health, and neurology and mental disorders. For stomatology, the clinical part includes general stomatology, periodontology, pediatric stomatology, oral and maxillofacial surgery, orthodontics, prosthodontics, and public health (Afghanistan Medical Council, 2022).

The administration of the medical Exit Exam has changed over time. As described by Rahimi et al. (2026), the Afghanistan Medical Council (AMC)

historically arranged these examinations for university and higher-education graduates, while the Ministry of Public Health (MoPH) and the National Examination Authority (NExA) were connected with examinations for health-science institute graduates (Rahimi et al., 2026). More recent official reporting from the MoPH stated that the ministry, in coordination with NExA, conducted the second round of the 1404 medical licensure examination for curative medicine and stomatology graduates (Ministry of Public Health, 2026). For this reason, this thesis has treated the Exit Exam as a national licensing examination whose administrative pathway has shifted over time, while keeping the project focus on the preparation needs of its candidates.

It is important to place the Exit Exam within the wider Afghan health examination landscape. The medical Exit Exam or NMLE is not the same as the health-science institute examinations taken by graduates of institute-level programs such as pharmacy, nursing, medical technology, anesthesia, and other related programs. It is also different from postgraduate specialization and residency examinations, which are connected to entry into, or completion of, specialty training after medical graduation. According to Rahimi et al. (2026), health-science institute examinations and postgraduate specialty pathways have different candidate groups, purposes, and administrative arrangements. Therefore, this project focused specifically on the licensing exam for curative medicine and stomatology graduates, not on institute-level health examinations or postgraduate specialization exams.

The structure of the Exit Exam makes it different from an ordinary semester test. It asks graduates to review a large part of their medical training and to perform under national, timed, high-stakes conditions. In principle, such an exam can support quality assurance by setting a common professional threshold across public and private institutions. In practice, preparation is difficult because Afghan medical education still faces serious resource and quality challenges. A recent commentary on medical education in Afghanistan reported weaknesses in infrastructure, learning environment, faculty capacity, access to updated educational resources, technology use, and financing (Stanikzai et al., 2025). The same paper noted that many programs still depended heavily on lecture notes and that available educational resources were not always aligned with Exit Exam requirements, leaving students to prepare for the exam largely on their own (Stanikzai et al., 2025).

Student-level evidence pointed in the same direction. In a cross-sectional study of senior Afghan medical students, only about one fifth of respondents rated their medical training as excellent, while many reported concerns related to session

organization, faculty support, and access to textbooks and journals (Stanikzai et al., 2023). These issues matter for Exit Exam preparation because the exam depends not only on remembering isolated facts, but also on regular revision, broad subject coverage, and confidence with medical multiple-choice questions. According to Parker (2020), repeated low-stakes quizzes can support long-term recall better than relying on rereading alone, especially when students are allowed to practice more than once and receive feedback. Similarly, according to Zhang et al. (2023), realistic practice exams have helped students recognize gaps in their knowledge and prepare more actively, although their effectiveness has depended on how the practice tasks and feedback are designed.

The Exit 360 app was proposed in response to this preparation need. The project focused on a mobile-first application that helps medical graduates study through organized question practice, timed mock tests, feedback, subject-wise revision, and progress tracking, with online weekly tests, leaderboards, and peer challenges available when internet access allows. It did not claim to solve the wider problems of medical education in Afghanistan. Instead, it addressed a smaller but practical gap by giving graduates a structured, accessible, and exam-oriented way to prepare for a national licensing exam that covers many subjects and has direct consequences for their professional future.

1.2 Problem Statement

According to Zhang et al. (2023), practice exams have been shown to improve exam preparation by helping learners discover knowledge gaps, practice actively, and retain material for a longer period. As emphasized by Parker (2020), repeated low-stakes testing has also been valuable, especially when learners have been able to try questions more than once and use feedback to guide further study. These findings were relevant for Exit Exam preparation because the exam has required broad, timed, multiple-choice practice across paraclinical and clinical subjects.

The main problem has been that many Afghan medical graduates have been expected to prepare for this licensing exam without enough structured, exam-oriented support. As described by Rahimi et al. (2026), the Exit Exam has been described as a national licensing assessment that covers a wide range of medical subjects and has had to be passed before graduates could obtain a medical practice permit. Therefore, preparation problems have not only affected exam performance. Unsuccessful or delayed success has also had the potential to postpone a graduate's entry into formal medical practice and delay the professional income expected after many years of study. A graduate

preparing for such an exam has needed more than general reading. They have needed a clear way to revise subject areas, attempt timed questions, review mistakes, and understand whether their preparation has been improving.

However, the available preparation environment has not fully supported this kind of study. According to Stanikzai et al. (2025), many medical programs in Afghanistan have had limited access to updated learning resources and have made weak use of educational technology. The same source also reported that available learning materials have not always been aligned with Exit Exam requirements. This has left graduates to connect the exam blueprint, textbooks, lecture notes, and practice questions mostly by themselves. In a separate study by Stanikzai et al. (2023), senior medical students reported concerns about session organization, faculty support, and access to textbooks and journals. These issues have made preparation feel scattered, especially when graduates have needed regular practice and timely feedback rather than only passive review.

Online learning has appeared to offer a useful path by reducing dependence on fixed class schedules and physical preparation centers. It has also made question practice, revision, and feedback more flexible for graduates who have had to study at different times and in different places. Still, a purely online solution has not been enough in the Afghan context. In a study by Gawhari et al. (2021), online education in Afghanistan was found to be affected by limited internet access, slow connection speed, high internet costs, and low availability of quality digital teaching materials.

Therefore, Exit Exam preparation has needed a solution that is structured but also realistic for local conditions. Graduates have needed access to subject-wise questions, timed mock tests, progress tracking, and feedback in a format that could continue working even when the internet was limited. At the same time, an internet connection could add further value through online weekly tests, leaderboards, and peer challenges that have encouraged regular practice. This has created the need for an offline-capable, mobile-first Exit Exam preparation application that would support steady practice while also offering online competitive features when connectivity was available.

1.3 Objectives

This section presents the main objective of the project and the sub objectives used to guide its design, implementation, and evaluation. The main objective states the overall purpose of developing an Afghan medical Exit Exam preparation mobile application, while the sub objectives divide this purpose into smaller

and more focused parts.

1.3.1 Main Objective

- To design and implement an Afghan medical Exit Exam preparation mobile application.

1.3.2 Sub Objectives

- To identify the existing challenges faced by Exit Exam candidates and analyze their main preparation needs.
- To design and implement an Exit Exam preparation mobile application based on the identified requirements.
- To test and evaluate the developed system in relation to its intended objectives.

1.4 Research Questions

This section presents the research questions that guide the study. The main research question is aligned with the main objective, while the sub research questions correspond to the sub objectives and help organize the study around problem identification, system development, and evaluation.

1.4.1 Main Research Question

- How can an Afghan Exit Exam preparation mobile application be designed and implemented?

1.4.2 Sub Research Questions

- What are the existing challenges faced by Exit Exam candidates and their main preparation needs?
- How can an Exit Exam preparation mobile application be designed and implemented based on the identified requirements?
- How can the developed system be tested and evaluated in relation to its intended objectives?

1.5 Scope and Limitations

This section defines what was included in the project and what was left outside its boundaries. The scope clarifies the main features, platforms, users, and data covered by the system, while the limitations describe the contextual, content, technical, and resource constraints that shaped the project.

1.5.1 Scope (Inclusions)

Features: The mobile application included a local question bank, offline subject-wise practice tests, general mixed practice tests, optional timed sessions, automatic grading, answer review, attempt history, and basic performance statistics. These features support repeated practice, progress awareness, and preparation across the clinical and paraclinical subjects of the Exit Exam. The online scope included authenticated user accounts, backend-managed weekly tests, leaderboard snapshots, peer or friend challenge workflows, secure server-side grading and basic notification support. The web admin panel included controlled tools for reviewing platform activity, inspecting question-bank summaries, planning weekly tests, publishing leaderboards, and triggering protected backend operations.

Platforms and technologies: The system was designed as an offline-capable Android application developed with Flutter. Local SQLite storage was included for the bundled question bank and offline attempt records, so that the main practice workflow can remain usable without an internet connection. The online side used Firebase services for account-based and competitive workflows. The backend scope included Firebase Authentication, Cloud Firestore, TypeScript Firebase Cloud Functions, and Firebase Cloud Messaging. The internal admin panel intended for controlled operational work was developed as a web application using React, TypeScript, and Vite.

Users: The target users are curative medicine and stomatology graduates preparing for the Afghan medical Exit Exam, as well as final-year students approaching the licensing stage.

Data: The data scope included question-bank content, local and online test attempts, performance statistics, user profile data, friendship and peer-interaction records, server-graded submissions, weekly test records, leaderboard records, challenge records, admin workflow data, and questionnaire responses used for requirements elicitation.

1.5.2 Limitations (Exclusions)

Context: The project focused on the Afghan medical Exit Exam context and did not aim to cover all health-sector examinations in Afghanistan. Health-science institute examinations, specialization or residency examinations, and other professional tests are outside the direct scope. The questionnaire used a limited sample of relevant respondents, so its findings should be treated as requirements-oriented evidence rather than a complete national measurement of all Exit Exam candidates.

Content: The question bank used in the project was limited in language and coverage, with the current content prepared in Dari. A complete, expert-verified, bilingual Dari and Pashto question bank is outside the current capstone scope because it requires medical expert review, careful content authorization, and long-term content maintenance. The system architecture supported future question-bank expansion, but the present work does not claim to provide a full official replacement for medical textbooks, preparation centers, or authorized exam materials.

Technical: Several advanced technical features were also excluded from the current scope. These include real-time challenge sessions, continuously updating live leaderboards, large-scale tournaments, advanced adaptive learning, video lessons, medical animations, real-time chat, and full production deployment to app stores. Challenge tests were treated as asynchronous workflows, and leaderboard results were published as backend-generated snapshots after grading to keep backend reads, writes, and operating costs manageable within the prototype scope.

Resource: The project was constrained by the capstone timeline, limited team size, Firebase usage costs, limited formal effectiveness testing, and the need to validate the system within a prototype environment. Therefore, the project focused on functional correctness, usability, offline access, secure backend-controlled online workflows, and preparation support rather than proving long-term improvement in official Exit Exam pass rates.

1.6 Significance of the Project

According to Rahimi et al. (2026), the Exit Exam has functioned as a required licensing assessment for medical and stomatology graduates who intend to work in Afghanistan. This has given the project practical significance because preparation for the exam has been connected not only to a single test result, but also to professional registration, work permission, and the transition from

medical education into practice. A graduate who has struggled to prepare effectively has faced the risk of delayed entry into formal medical work after many years of study. Supporting preparation for this exam has therefore been an important educational and professional need.

Exit 360 was developed to address this need by providing a structured mobile preparation system for the Afghan Exit Exam. The system focused on subject-wise practice, timed mock tests, automatic feedback, progress tracking, and review of previous attempts. These features have been significant because they turned preparation from scattered reading into a more organized process of practice, review, and improvement. As emphasized by Parker (2020), repeated low-stakes testing has supported learning when students have been able to practice more than once and use feedback to guide further study. Similarly, according to Zhang et al. (2023), realistic practice exams have helped learners identify knowledge gaps and prepare more actively. Research on mobile quiz applications has also suggested that learners have benefited from being able to test themselves at flexible times and places (Rüth et al., 2021). Other studies have reported that on-demand mobile assessments have increased practice opportunities, engagement, and learning satisfaction (Lazarinis et al., 2017).

The project has also been significant because it was designed for local access conditions. Studies on Afghan medical education have reported limited access to updated learning resources, weak use of educational technology, and learning materials that have not always been aligned with Exit Exam requirements (Stanikzai et al., 2025). Student-level evidence has also shown concerns related to session organization, faculty support, and access to textbooks and journals (Stanikzai et al., 2023). By making the core question bank, practice tests, results, and progress history available offline, Exit 360 has been able to support graduates who have not been able to depend on constant internet access or regular attendance at physical preparation centers.

At the same time, the online features have added further value when connectivity was available. Weekly tests, leaderboards, and peer challenges have been able to encourage regular practice and create a sense of learning momentum. In a review by Li et al. (2024), leaderboards in higher education were found to support motivation, engagement, and performance. According to Abdul Halim et al. (2022), leaderboard-based gamification has also helped improve participation and encourage learning beyond regular class activity. According to Ma et al. (2024), healthy academic competition has helped encourage learners to invest effort in their studies.

Overall, Exit 360 has been significant because it combined structured exam

practice with realistic access design. It supported offline preparation as the foundation, while using online activities as an additional motivational layer. In this way, the project responded to the specific preparation needs of Afghan medical graduates and offered a practical tool for improving readiness for a national licensing exam with direct consequences for their professional future.

1.7 Research Methodology

This study used a quantitative research methodology to support requirements elicitation for an Afghan medical Exit Exam preparation mobile application. Before preparing the questionnaire, an initial feature set was derived from the literature review and the analysis of existing local and foreign medical exam preparation systems. This helped identify relevant preparation features such as question banks, timed practice tests, answer review, progress tracking, offline access, and online learning activities.

The main data collection instrument was a structured questionnaire administered to medical graduates and final-year students preparing for the Afghan medical Exit Exam. The questionnaire collected feedback about preparation habits, access to practice tests, internet availability, offline-access needs, feature usefulness, feature priorities, and expected impact. Although Cochran's formula indicated that a larger sample would be required for a statistically representative survey, the collected responses were treated as requirements-oriented evidence rather than nationwide statistics.

The questionnaire responses were analyzed using descriptive statistics such as frequencies, percentages, mean scores, and weighted priority rankings. The results helped summarize the main preparation problems faced by Exit Exam candidates and identify which proposed features were considered most useful. These findings were then used to refine requirements in areas such as offline question practice, timed mock tests, answer review, progress tracking, synchronization, and optional online competition.

1.8 Development Methodology

The development of Exit 360 followed an Agile development methodology with a lightweight Scrum-based organization. The project was divided into manageable iterations rather than being implemented as one fixed sequence of tasks. This approach was suitable because the system combines an offline mobile preparation experience, local storage, Firebase-backed online services, and an internal web admin panel. It also allowed the team to review and

improve each main part of the system as requirements became clearer through the literature review, questionnaire findings, prototype review, and supervisor feedback.

The first iteration focused on the offline Flutter foundation, including navigation, local Exit Exam question storage, subject-wise and mixed practice tests, timers, local grading, result summaries, answer review, attempt history, and basic statistics. The second iteration added the online and social layer through Firebase Authentication, Firestore, Cloud Functions, user profiles, friendship, weekly tests, leaderboards, asynchronous challenges, synchronization, notifications, and security-rule validation. The third iteration focused on the React, TypeScript, and Vite admin panel for controlled operational workflows such as question-bank management; weekly-test management and leaderboard publishing; read-only summaries for users, sessions, submissions, and challenges; and platform inspection.

Each major iteration produced a prototype that was reviewed before the next stage of work. The first prototype focused on offline preparation, the second on online social and competition features, and the third on admin and operational workflows. Git and GitHub were used throughout the monorepo, while Flutter tooling, Android Studio, Visual Studio Code, Firebase tooling, and Chrome DevTools supported implementation, testing, backend validation, and documentation.

1.9 Outcomes

The main outcome of the project was Exit 360, an Afghan medical Exit Exam preparation system designed around offline access, structured practice, progress feedback, and optional online competition. The implemented work included a Flutter and Dart mobile application with a local SQLite question bank, subject-wise and mixed practice tests, timed sessions, automatic grading, answer review, attempt history, and basic statistics.

The online layer included Firebase Authentication, Firestore data flows, Cloud Functions for trusted competitive workflows, user accounts, weekly tests, leaderboard-related logic, friend and challenge operations, notification support, and backend validation through the Firebase Emulator Suite and deployed function testing. The project also produced an internal React, TypeScript, and Vite admin panel for controlled operational workflows such as question-bank management, weekly-test management, leaderboard publishing, and platform inspection.

In addition to the working software, the project produced system design artifacts such as use case diagrams, sequence diagrams, architecture design, entity-relationship design, and relational schema design. It also produced questionnaire-based requirements evidence, prototype review findings, and an automated Flutter test suite covering important domain logic, models, controllers, and selected widget flows. Together, these outcomes documented the requirements, design, implementation, testing, scope, limitations, and planned improvements of the system, all of which were packaged in the current LaTeX thesis.

1.10 Organization of the Report

- Chapter 1 introduced the Afghan medical Exit Exam preparation problem, the background of the project, the research objectives, the research questions, the scope and limitations, and the significance of the proposed mobile preparation application. It also provided brief summaries of the research methodology, development methodology, project outcomes, and the overall organization of the report.

- Chapter 2 reviewed related work and existing systems in the area of medical licensing exam preparation. It discussed local Afghan Exit Exam preparation applications, selected international medical exam preparation platforms, recent educational technology trends, and the research gap that the project aimed to address.

- Chapter 3 presented the methodology of the project. It explained the research methodology, requirements elicitation, questionnaire design, population and sampling, data analysis method, functional and non-functional requirements, development methodology, system design, implementation, testing, and the main challenges and solutions encountered during the project.

- Chapter 4 presented the results of the questionnaire analysis. It summarized the respondent profile, current practice and access conditions, internet and offline-access needs, feature usefulness and priorities, likelihood of adoption, perceived impact, and the way these findings support the first research objective and research question.

- Chapter 5 discussed the meaning of the results and implementation findings in relation to the research questions and project objectives. It interpreted how the survey findings, system design decisions,

implementation choices, and testing outcomes support the overall argument of the thesis.

- Chapter 6 concluded the thesis by summarizing the work completed, the main contributions of the project, the achievement of objectives, and the limitations that remain. It also outlined future enhancements such as expanding the question bank, adding stronger bilingual content support, improving online features, conducting more formal user evaluation, and preparing the system for wider deployment.

CHAPTER 2: LITERATURE REVIEW

This chapter reviews the existing systems, technologies, and research findings that informed the design of Exit 360. It first examines local Afghan medical exam preparation applications, then reviews selected international medical licensing exam platforms for design lessons relevant to the Afghan Exit Exam context. The chapter also discusses recent educational and technological trends related to mobile practice, offline access, analytics, and competitive learning. It concludes with a synthesis of the reviewed literature and explains how its practical lessons guided the locally focused design of Exit 360.

2.1 Related Work and Existing Systems

Medical exam preparation apps have become an important part of contemporary test preparation worldwide. Therefore, this section reviewed existing digital systems for medical licensing and high-stakes examination preparation in order to position Exit 360 within both the Afghan Exit Exam context and the broader landscape of exam-preparation technologies. Since the Afghan Exit Exam is a national licensing requirement that affects graduates' registration and entry into formal medical practice (Rahimi et al., 2026), the review first considered locally relevant Afghan medical exam preparation applications rather than beginning with general international platforms. The purpose of the review was not simply to list available applications, but to examine how existing systems support core preparation needs such as access to question banks, timed practice, automatic grading, feedback, progress monitoring, offline use, synchronization, and competitive learning. After the local systems were examined, selected comparable medical exam preparation platforms from other contexts were discussed only where their mechanisms provide useful design insight for Exit 360. This analytical approach helped identify the practical and technical gaps that remain in current solutions, especially for Afghan medical graduates who may face limited connectivity, unequal access to preparation resources, and limited structured support while preparing for the national licensing exam.

2.1.1 Local Medical Exam Preparation Applications

Within Afghanistan, the current mobile preparation ecosystem for the Exit Exam is still narrow. The main locally oriented applications identified for this review were two Android apps, MedQuiz and EXIT Medical Exam Prep, each attempting to support health-science graduates through mobile-based question practice. Firstly, MedQuiz, locally known as Azmoonak Tebi, is offered by the M. Saber Sabiry Institute of Health Science (SIHC), and is relevant because

it includes question packs for the Exit Exam for both curative medicine and stomatology graduates (Azmoonak Tebi, 2026). However, the application is not dedicated only to the Exit Exam since it also presents question packs for other health-science institute examinations, including nursing, pharmacy, dental prosthesis, medical technology or laboratory sciences, and midwifery. MedQuiz appears to follow an online-first model because users must log in or sign up before accessing the main content. At startup, it provides only ten static demo questions, while the remaining question packs are placed behind paid, limited-time subscriptions. Because the app is new and has no public review base, the size, quality, and exam alignment of its paid question packs cannot be judged from public information alone. The available screenshots suggest that the app includes test results, answer review, attempt history, basic statistics, and performance tracking, which are useful elements for exam preparation. Compared with Exit 360, however, MedQuiz appears more limited in offline access and transparency for new users. Exit 360 was designed around an offline-capable core question bank, timed mock tests, progress tracking, and review features, while using online weekly tests, leaderboards, and peer challenges as additional motivational features when connectivity is available. Therefore, MedQuiz shows that there is local interest in mobile medical exam preparation, but it also highlights the need for a more focused, accessible, and offline-capable Exit Exam preparation system.

Secondly, EXIT Medical Exam Prep by Zahid apps store represents a stronger offline model for local medical exam preparation (Zahid apps store, 2026). The application appears to provide a large bilingual Dari and Pashto question bank immediately after installation, without requiring any login, account creation, or payment. This is a notable strength in the Afghan context because it allows graduates to begin practicing without depending on internet access, registration, or a paid entry step. For the Exit Exam itself, the app includes two complete 160-question papers from 1401 and 1405, which can help users become familiar with full exam length and question style; however, its Exit Exam coverage is limited to only those two papers and it does not offer comprehensive subject-wise practice based on the official exam blueprint for each of the paraclinical and clinical subjects in curative medicine and stomatology. However, like MedQuiz, the app also includes practice questions for other health-science institute examinations, specifically medical technology, pharmacy, and dental prosthesis, this time with proper subject breakdown. The application supports timed tests, results, and answer review, which are important baseline features for exam practice. Subject-wise practice for each exam remains fully offline and unlimited, but creating a general practice test for a selected exam seems to have a daily limit and eventually requires watching an advertisement, which depends

on internet access. Nevertheless, the app appears to offer little beyond the immediate test session. It records the best percentage for a specific subject test, but there is no visible support for attempt history, detailed statistics, progress tracking, or performance analysis across subjects and repeated attempts. It also has no online learning layer, such as weekly tests, leaderboards, or peer challenges. Compared with Exit 360, EXIT Medical Exam Prep is strong in offline access and direct availability of bilingual questions, but as of now, it remains limited as a dedicated long-term preparation system for the Exit Exam. Exit 360 aimed to preserve the offline advantage while adding alignment with the official Exit Exam structure and subject breakdown, progress monitoring, practice history, and optional online competitive features when internet access is available.

It is also important to note that both local applications are very recent. MedQuiz was released on January 18, 2026, had more than 100 downloads, no public rating or reviews, and was listed as version 1.0.1 at the time of this review (Azmoonak Tebi, 2026). EXIT Medical Exam Prep was released on April 15, 2026, had more than 500 downloads, a 4.4 rating based on only seven reviews, and was listed as version 1.1.0 (Zahid apps store, 2026). Therefore, their presence does not remove the need for Exit 360, especially because both apps have appeared months after the project identification and selection phase. It also shows that the local app landscape is changing quickly. Newer development approaches, including AI-assisted and agentic coding, suggest that small teams can release and update apps faster than before. For this reason, the comparison above should be understood as a snapshot of these applications as they appeared in June 2026, not as a final judgment of how they may develop in the future.

2.1.2 International Medical Licensing Exam Preparation Systems

After the local mobile preparation landscape was examined, the review moved to foreign medical licensing preparation systems. These systems are not direct competitors to Exit 360 because they are built for different examinations, languages, institutions, and markets. However, they are useful design references because they show how mature exam-preparation products organize large question banks, timed mocks, answer review, progress tracking, revision support, and optional online services. The following applications were therefore reviewed as selective design references for an Afghan Exit Exam preparation system, not as models to copy without adjustment.

UWorld Medical Exam Prep is one of the most recognized preparation systems for the United States Medical Licensing Examination (USMLE), including Step

1, Step 2 Clinical Knowledge (Step 2 CK), and Step 3 (UWorld, 2026). These steps represent different stages of the USMLE pathway, from basic science foundations to clinical knowledge and later readiness for unsupervised medical practice. It is available through iOS, Android, and web platforms, which allows candidates to move between mobile and desktop study. Its core value is a large exam-style question bank combined with custom quizzes, self-assessments, answer explanations, smart flashcards, spaced repetition, a medical library, videos, study planning, offline study, and synchronization across devices. For Exit 360, UWorld shows the value of treating practice as a complete study cycle rather than as isolated question solving. Its flashcards, videos, and large commercial content library are beyond the current scope of Exit 360, but its emphasis on timed blocks, review, weak-area awareness, and offline continuation is directly relevant to Afghan graduates who need steady preparation even when connectivity is unreliable.

AMBOSS is a global medical knowledge and question-bank ecosystem used for examinations such as USMLE, United States medical school Shelf exams, the Medical Council of Canada Qualifying Examination (MCCQE), the Saudi Medical Licensing Exam (SMLE), and the International Foundations of Medicine (IFOM) (AMBOSS, 2026). It is available on iOS, Android, and web platforms, with mobile access extending the main knowledge and question-bank experience. Its major strength is the close connection between a searchable medical library and board-style questions, supported by personalized recommendations, score prediction, image overlays, Anki integration, study plans, and an offline knowledge app. This model is valuable because it connects practice questions to explanation and reference material. For Exit 360, the relevant lesson is not to reproduce a full medical encyclopedia, which would be too large for the project scope, but to connect each practice attempt with review, explanation, subject mapping, and progress signals that help graduates understand what to revise next.

Plabable is focused on the Professional and Linguistic Assessments Board (PLAB) examinations in the United Kingdom (Plabable, 2026). It is offered across iOS, Android, and web platforms, making it accessible as both a mobile and browser-based study tool. It provides a large set of high-yield PLAB questions, mock exams, revision guides, flashcard-style Gems, comment threads, and community support through dedicated groups. Its design is narrower than broader knowledge platforms such as AMBOSS, but that narrowness is also its strength because the content is clearly framed around one licensing route. For Exit 360, Plabable suggests that a focused exam-specific system can be useful even without building a very large multimedia learning platform. However, its

community and group-based support depends on online participation, while Exit 360 must keep the core practice experience usable offline and make social or competitive features optional.

Quesmed supports the United Kingdom Medical Licensing Assessment (UKMLA), PLAB, Membership of the Royal Colleges of Physicians (MRCP), Multi-Specialty Recruitment Assessment (MSRA), and other examinations in the United Kingdom (Quesmed, 2026). It is available through iOS, Android, and web platforms, which supports both phone-based revision and longer study sessions on larger screens. It combines a question bank with integrated notes, progress trackers, detailed filters, dedicated test mode, offline downloads for selected questions, image-based questions, flashcards, and clinical examination resources. This makes Quesmed a useful example of combining exam simulation with topic-level filtering and structured revision. For Exit 360, the most relevant ideas are subject filters, timed practice, answer review, and offline access to selected content. Its broader clinical examination video features are useful in medical education, but they are outside the current scope of an Exit Exam preparation app centered on multiple-choice practice.

PassMedicine is used for medical finals and UKMLA preparation in the United Kingdom (PassMedicine, 2026). It is delivered as an installable progressive web application (PWA), mobile web system, and desktop web system rather than as a conventional native-only app. It provides a large single-best-answer question bank, high-yield textbooks, extended notes, timed tests, mock exams, peer benchmarking, and a knowledge tutor that supports repeated review of facts. For Exit 360, PassMedicine shows how a preparation product can combine question practice, explanations, revision notes, and performance comparison without relying mainly on live teaching or heavy multimedia. The peer benchmarking idea is relevant to Exit 360's online layer, while the offline and installable approach reinforces the importance of flexible access in lower-connectivity contexts.

Marrow is a major Indian medical exam preparation platform for the National Eligibility cum Entrance Test for Postgraduate (NEET-PG), India's National Exit Test (NExT), and super-specialty examinations (Marrow, 2026). It is available on iOS, Android, and web platforms, which supports the large video and test ecosystem across devices. Its features include extensive video lessons, large multiple-choice question sets, incorrect-question retargeting through the Magic Module, topic mapping, many tests and grand tests, analytics, animations, and clinical scenarios. Marrow demonstrates how a mature medical preparation platform can combine teaching content, repeated practice, and

analytics at scale. For Exit 360, the most relevant parts are structured mock tests, incorrect-question review, subject mapping, and progress analytics. Its large video library, animations, and clinical scenarios are valuable but beyond the intended scope of this capstone, which focuses more narrowly on offline-capable practice and feedback.

PrepLadder is another Indian platform for NEET-PG, the Institute of National Importance Combined Entrance Test (INI-CET), the Foreign Medical Graduate Examination (FMGE), and NEET super-specialty examinations (PrepLadder, 2026). It is offered through iOS, Android, and web platforms, which fits its role as a multi-device coaching and revision system. It combines video lectures, a high-yield question bank, detailed analytics, three-dimensional (3D) models, animations, real patient cases, rapid revision content, and material mapped to past questions and current exam patterns. This shows the value of building a preparation workflow around both content delivery and repeated testing. For Exit 360, the key design lesson is exam-pattern alignment and useful analytics, not the replication of expensive multimedia production. In the Afghan context, a lighter system with a reliable local question bank, timed mock tests, and clear subject-based progress may be more realistic than a large video-centered coaching platform for now.

DAMS, or Delhi Academy of Medical Sciences, is another Indian medical exam preparation ecosystem for NEET-PG, FMGE, INI-CET, National Eligibility cum Entrance Test for Master of Dental Surgery (NEET-MDS), and related examinations (DAMS, 2026). It is available through iOS, Android, and web platforms, matching its blend of app-based practice and online coaching. Its feature set includes live two-way interactive classes, recorded courses, multi-level testing, a question bank with video solutions, rapid revision, previous-year-question focus, and AI-enabled competitive assessment. DAMS is therefore especially strong as a coaching and live-learning ecosystem. For Exit 360, however, live classes, video solutions, and broad coaching services are outside scope because they require large teaching teams, strong connectivity, and continuous content production. The useful design lessons are the testing ladder, repeated assessment, competitive motivation, and focus on exam-style practice, which can be adapted through weekly tests, leaderboards, and peer challenges when internet access is available.

CanadaQBank is a Canada-focused but multi-exam platform that supports the MCCQE and also offers resources for other exams such as the Pharmacy Examining Board of Canada (PEBC), Australian Medical Council (AMC), PLAB, SMLE, and USMLE (CanadaQBank, 2026). It is available on iOS,

Android, and web platforms. Its MCCQE Part I preparation includes thousands of multiple-choice questions, timed and untimed modes, tutor mode, category filters, notes, subject analytics, comparison with other users, and synchronization between desktop and mobile use. This makes it relevant to Exit 360 because it treats exam preparation as repeated practice across subjects and categories, not only as full mock tests. Its user comparison and analytics are useful references for Exit 360's progress and online competition ideas, but the Afghan system must remain simpler, localized, and usable without constant account-based synchronization.

The Australian Medical Council Multiple-Choice Question (AMC MCQ) Preparation App, delivered through eMedici, is a notable case because it is connected to an official licensing body rather than being only a commercial vendor product (Australian Medical Council, 2026). Through the eMedici environment, it provides desktop and mobile access on iOS and Android, along with AMC-written practice questions, immediate feedback, and periodic refresh of content. The broader eMedici environment also includes larger question banks, mock exams, case studies, and Objective Structured Clinical Examination (OSCE) resources. For Exit 360, the most important lesson is the value of official-style exam alignment. Even though Exit 360 is not an official board app, its credibility depends on aligning questions, subjects, and mock tests with the Afghan Exit Exam structure and subject distribution rather than simply collecting general medical questions.

PassGP serves Australian and New Zealand medical examination pathways, including AMC, Royal Australian College of General Practitioners (RACGP), Australian College of Rural and Remote Medicine (ACRRM), Royal Australian and New Zealand College of Obstetricians and Gynaecologists (RANZCOG), and related exams (PassGP, 2026). It is available through iOS and web platforms, linking app-based study with desktop access. It provides a large question collection, full mock exams, an AI virtual tutor, flashcards, structured tutorials, academy content, and integration between app and desktop study. Its broad multi-exam coverage is useful for users in several pathways, but it also requires a content and support model much larger than Exit 360. For the Afghan context, the practical lessons are full mock exams, flashcards or review tools, and guided feedback. The AI tutor and broad academy content are interesting future directions, but this project focused first on reliable offline practice and measurable progress for one licensing exam.

SMLE Gate is focused on the Saudi Medical Licensing Exam in Saudi Arabia (SMLE Gate, 2026). It is available as a web platform with iOS and Android app

access, which makes it more comparable to the broader multi-platform systems reviewed above. It offers simulated exams with real timing, a comprehensive question bank, performance tracking, result review, notes, saved questions, and a mobile interface designed for focused exam preparation. Compared with larger ecosystems such as UWorld, AMBOSS, or Marrow, SMLE Gate appears closer to the type of focused licensing-exam app that Exit 360 aimed to be. Its emphasis on simulated exams, review, notes, and performance tracking is directly relevant, while its country-specific content shows why Exit 360 must be localized around the Afghan Exit Exam rather than relying on general international medical question banks.

Overall, the global review showed that modern medical exam preparation systems are no longer just simple question lists. The strongest products combine exam-aligned question banks, timed practice, answer review, analytics, structured revision, weak-area targeting, and in some cases multimedia teaching or AI support. Many of these advanced features depend on stable internet, large content teams, subscription ecosystems, or official board access, which cannot be assumed for Afghan graduates. Another repeated pattern is broad platform support, since most mature systems combine mobile apps with web access so that candidates can move between phone-based and desktop study. Exit 360 was first planned as an Android application, but its Flutter-based implementation can make future expansion to additional platforms more practical. Therefore, Exit 360 selectively adapted the practical parts of these systems: subject-wise practice, timed mock tests, review, progress tracking, offline access, and optional online competition. Its design was kept locally focused, offline-capable, and aligned with the official Exit Exam subject breakdown.

Taken together, the local and international systems represented different levels of maturity and different approaches to medical exam preparation. The local applications reflected Afghan language, access, and examination needs, while the international systems provided stronger examples of structured practice, answer review, progress tracking, analytics, and competitive motivation. Many advanced international features, however, depended on large content teams, stable connectivity, official content access, or commercial infrastructure. These findings supported a selective approach for Exit 360, which adapted the most practical preparation features while remaining offline-capable, locally focused, and aligned with the Afghan Medical Exit Exam context.

2.2 Recent Trends and Technologies

Recent developments in educational technology show a clear shift from static digital content toward interactive, data-informed, and mobile-first learning systems. In the context of high-stakes licensing examinations, this shift is especially important because candidates do not only need access to reading material; they also need repeated practice, feedback, progress awareness, motivation, and a clear way to organize their preparation. Exit 360 was therefore connected to several related trends: mobile self-regulated learning, offline-first design, cloud-backed synchronization, learning analytics, gamified competition, and cross-platform development. These trends are not independent from one another. In a preparation system, they work together: mobile access allows frequent practice, offline storage protects continuity, cloud services support shared records and competition, and analytics convert attempts into practical feedback.

One of the most relevant trends is mobile learning, particularly through quiz and practice-test applications. Mobile devices allow learners to practice outside fixed classroom schedules, which is valuable for students who cannot regularly attend preparation centers or who need flexible study time. According to Rüth et al. (2021), mobile quiz applications can support self-regulated learning by giving learners flexibility in terms of location, time, and device. Similarly, according to Lazarinis et al. (2017), mobile self-assessment allows learners to test themselves at their own pace and receive feedback without depending on a formal classroom session. For Exit 360, this trend directly supported the decision to build an Android-focused mobile application rather than only a web-based system. Medical graduates preparing for the Exit Exam often need short, repeated practice sessions across many subjects, and mobile access makes this more realistic than a system tied to a specific place or schedule.

The educational value of repeated testing is another important foundation for modern exam preparation systems. Research on practice exams shows that testing is not only a way to measure knowledge, but also a learning activity in itself. As argued by Parker (2020), repeated low-stakes quizzes can support long-term learning more effectively than a single high-stakes assessment, while according to Zhang et al. (2023), realistic and earlier practice exams can influence students' metacognition, study behavior, and exam performance. In the Exit Exam context, this supports the use of timed subject tests, general mock tests, and weekly exams. A preparation application should not simply store past questions; it should create conditions where students can repeatedly practice under exam-like constraints, receive quick feedback, and adjust their

study behavior over time.

Because Exit 360 was designed for Afghan medical graduates, offline-first architecture is a particularly important technological trend. Online learning in Afghanistan faces practical barriers such as limited connectivity, slow internet speed, high internet costs, and uneven availability of quality digital learning resources (Gawhari et al., 2021). A purely online preparation system may therefore fail exactly when students need it most. Offline-first design responds to this problem by placing essential learning resources on the device and treating the network as an enhancement rather than a permanent requirement. In Exit 360, this meant storing a prebundled question bank locally and allowing students to take subject-wise and general tests even without internet access. SQLite is suitable for this kind of embedded local storage because it is a small, self-contained, full-featured SQL database engine widely used in mobile and desktop applications (SQLite Consortium, 2026). This supports the system's need for fast local access to questions, options, answers, and attempt records.

At the same time, offline-only systems can become limited if users cannot preserve records, compare performance, or participate in shared activities. For this reason, another recent trend is the combination of local storage with cloud-backed synchronization. Cloud services allow a system to manage authenticated accounts, shared tests, remote grading, leaderboards, and cross-device or restored-connectivity workflows. Firestore documentation, for example, describes offline persistence in which actively used data can be cached locally and local changes can be synchronized with the backend when the device comes back online (Google Firebase, 2026). Although Exit 360 used its own project-specific data model and synchronization rules, the same general principle is important: students should be able to continue learning offline while the system later uploads attempts, updates online records, and participates in competitive workflows when internet access becomes available. This is especially relevant for weekly tests and challenge submissions, where authoritative server-side records are needed for fairness and consistency.

Learning analytics is another major trend in educational applications. Instead of only telling a student whether a single answer is correct or wrong, analytics can summarize performance across attempts, subjects, time periods, and test types. At the application level, Exit 360 applied this idea by tracking individual attempts, scores, and progress indicators. Such analytics help students see whether they are improving, which subjects require more work, and how their performance changes across practice sessions. This addresses a limitation observed in many existing local apps, where feedback often stops at the number

of correct and wrong answers after a single test.

Gamification and competitive learning have also become common in exam-preparation systems. The purpose of gamification in this context is not entertainment for its own sake, but motivation, engagement, and repeated participation. Leaderboards are one of the most common competitive mechanisms. In a systematic review by Li et al. (2024), leaderboards in higher education were found to positively influence learning motivation, engagement, and performance. According to Abdul Halim et al. (2022), leaderboard gamification can also improve participation and encourage extra learning activity. For Exit 360, leaderboards are relevant because Exit Exam preparation is also high-stakes and competitive; students often want to know how their performance compares with others. However, the implementation must match the local context. Instead of real-time leaderboard streaming, which may increase bandwidth and backend cost, Exit 360 used snapshot-based leaderboard updates for weekly tests. This preserved the motivational value of ranking while keeping the system more practical for low-connectivity environments.

Related to gamification is the trend of peer-based and challenge-based learning. Competition can increase effort when it is structured in a healthy way. According to Ma et al. (2024), competitive attitude and a supportive competitive environment can be associated with stronger task motivation and performance. In Exit 360, this idea was implemented through the asynchronous challenge workflow, where students can challenge followers without needing a real-time session. This design is important because real-time multiplayer testing would be more difficult under unstable internet conditions. An asynchronous model allows students to compete, compare results, and remain socially engaged while still respecting the connectivity constraints identified in the Afghan educational context.

Finally, cross-platform and cloud-integrated development tools have made it more feasible for small student teams to build sophisticated educational systems within limited time and resource constraints. Flutter is relevant in this regard because it is a Google-supported framework for building applications across multiple screens and platforms from a shared codebase (Google, 2025). For Exit 360, Flutter supported the development of a mobile-first interface while leaving room for future expansion. Combined with local SQLite storage and cloud-backed services, this technology stack supported the main design goal of the project: an application that remains useful offline, becomes richer when online, and connects practice, review, competition, and analytics in one coherent workflow. Overall, the recent trends most relevant to Exit 360 are not simply

fashionable technologies; they are practical responses to the educational and infrastructural realities of Afghan Exit Exam preparation.

2.3 Literature Review Synthesis

The reviewed local applications showed that mobile preparation for the Afghan Exit Exam had already started to emerge, but the local landscape remained early and incomplete. MedQuiz demonstrated local interest in digital question packs for the Exit Exam and other health-science examinations, but its online-first access model, limited visible demo content, paid bank-based structure, and uncertain public evidence about question-bank depth left important gaps for students who need dependable daily practice (Azmoonak Tebi, 2026). EXIT Medical Exam Prep provided a stronger offline model with Dari and Pashto questions, timed tests, answer review, and two complete Exit papers, but it remained limited in long-term progress tracking, online competition, and broader Exit-specific coverage (Zahid apps store, 2026). These two systems were valuable signs of a growing local app landscape, but they did not fully address the need for a focused, transparent, offline-capable, and progress-aware Exit Exam preparation system.

The foreign systems reviewed in this chapter showed a more mature direction for medical licensing exam preparation. UWorld treated practice as a full cycle of timed questions, explanation, review, assessment, and weak-area awareness (UWorld, 2026). AMBOSS connected question practice with a larger medical knowledge base and performance signals (AMBOSS, 2026), while SMLE Gate showed how a national licensing-exam platform could focus on simulated exams, saved questions, notes, result review, and performance tracking (SMLE Gate, 2026). However, these systems were designed for other countries, languages, examination bodies, and market conditions. Their models often assume larger content teams, stronger subscription ecosystems, wider platform support, and more stable internet access than many Afghan graduates can reliably depend on.

The technology and learning trends discussed in this chapter also point to a specific design gap. According to Rüth et al. (2021), mobile quiz tools can support flexible, self-regulated study, and according to Lazarinis et al. (2017), mobile self-assessment allows learners to practice and receive feedback outside a formal classroom. As discussed by Parker (2020), repeated quizzes can support long-term learning, while according to Zhang et al. (2023), realistic practice exams can support students' study behavior and exam readiness. At the same time, Afghan online learning conditions made offline-first design important (Gawhari et al., 2021). Therefore, the gap was not simply the absence

of question apps. The stronger gap was the absence of an Exit Exam preparation system that combined exam-aligned practice, offline continuity, timed mock testing, answer review, progress analytics, and optional online competition in one coherent workflow.

Exit 360 was designed to respond to this gap by adapting the practical lessons from both local and foreign systems while remaining focused on the Afghan Exit Exam context. Its offline layer was intended to support subject-wise and general tests through a prebundled local question bank. Its online layer added authenticated profiles, synchronization, weekly tests, leaderboard snapshots, and asynchronous challenges when internet access was available. Its analytics features aimed to help students understand performance across attempts instead of only seeing the result of one test. In this way, the project did not claim that no Exit Exam preparation applications exist. Rather, it addressed the remaining gap between isolated question-practice tools and a more complete, offline-capable preparation system suited to Afghan medical graduates' learning and connectivity realities.

CHAPTER 3: METHODOLOGY

This chapter explains the research and development methods used to design and implement Exit 360. It describes the quantitative questionnaire-based requirements elicitation process, the target population and sampling approach, data collection and analysis, system requirements, and the main development methodology followed during the project. It also presents the system design, architecture, database design, implementation approach, and testing activities that connect the identified preparation needs with the final mobile application.

3.1 Research Methodology

This study used a quantitative research methodology to support the requirements elicitation for Exit 360. The main data collection instrument was a structured questionnaire administered to medical graduates and final-year students preparing for the Afghan Medical Exit Exam. It was used to measure candidates' preparation habits, access challenges, internet availability, feature priorities, and expected usefulness of an Exit Exam preparation mobile application.

Before preparing the questionnaire, an initial feature set was derived from the literature review and analysis of existing medical exam preparation systems both local and foreign. The reviewed literature helped identify the educational value of repeated practice, timed tests, feedback, and progress monitoring, while the existing-system review showed how local and international preparation apps support question banks, mock tests, answer review, offline access, analytics, and online learning features. These sources were used to create a preliminary list of features for Exit 360, which was then presented to the target users through the questionnaire.

The quantitative questionnaire responses were analyzed using descriptive statistics such as frequencies, percentages, mean scores, and priority rankings. This made it possible to summarize the main preparation problems faced by Exit Exam candidates and to identify which proposed features were considered most useful. The findings were then used to refine the functional and non-functional requirements of Exit 360, especially in areas such as offline question practice, timed mock tests, answer review, progress tracking, synchronization, and optional online competition.

3.1.1 Requirements Elicitation

Requirements elicitation began by translating the problem context into a practical feature list for Exit 360. The literature review showed that Exit Exam candidates

need repeated practice, exam-like testing, feedback, and support for broad subject revision. The review of local and foreign medical exam preparation applications then helped identify common system patterns, such as question banks, timed mock tests, answer review, progress tracking, offline content, and online learning activities. These sources provided the starting point for defining what an Exit Exam preparation application should support.

Based on this initial analysis, a preliminary set of proposed features was prepared for user validation. The proposed features included offline subject-wise practice, general practice tests, timed test sessions, automatic grading, answer review, attempt history, performance statistics, authenticated online profiles, synchronization, weekly tests, leaderboards, and challenge tests. The purpose of preparing this feature list was to make the requirements concrete enough for respondents to evaluate, rather than asking only general questions about whether a preparation application would be useful.

A structured questionnaire was then used to collect feedback from the target users. The questionnaire measured preparation habits, access to practice tests, internet availability, previous use of preparation applications, perceived usefulness of proposed features, feature priority rankings, and willingness to use an offline-capable Exit Exam preparation app. These responses helped determine which requirements were most important for the first version of Exit 360.

The final requirements were therefore shaped by both literature-based feature identification and quantitative user feedback. This helped keep the application focused on real preparation needs, including offline access, organized question practice, timed mock testing, review, progress awareness, and optional online competition when internet access is available.

3.1.2 Population and Sampling

The target population of this study consisted of individuals directly relevant to Afghan Medical Exit Exam preparation. This included curative medicine and stomatology graduates, as well as final-year curative medicine and stomatology students who were approaching graduation and preparing for the licensing exam. This population was selected because Exit 360 was intended for medical graduates and senior students who need support in subject-wise revision, practice testing, answer review, progress tracking, offline study, and optional online competition.

Although a broader population of all curative medicine and stomatology students

could have been considered, it was not selected because it would include many students in earlier academic years who are not yet actively preparing for the Exit Exam. Such respondents may have limited experience with the pressure, preparation habits, and practical needs of the licensing stage. Their answers could therefore be less useful for evaluating features such as timed mock tests, Exit-focused question banks, review of previous attempts, and weekly online tests. For this reason, the study prioritized respondents who were closer to the exam and more able to judge the proposed features from a realistic preparation perspective.

Graduates were included because they had already completed the medical or stomatology degree and were either preparing for the licensing exam or had direct experience with it. Final-year students were included because they represent the next group of likely candidates and are close enough to graduation to evaluate the usefulness of an Exit Exam preparation application. This combination helped balance retrospective experience from graduates with current preparation needs from senior students.

For sample size estimation, the main population size was represented by the annual number of graduates taking the medical licensure examination. The official MoPH report on the second round of the 1404 medical licensure exam states that around 7,000 curative medicine and stomatology graduates participated in the examination process (Ministry of Public Health, 2026). Although the questionnaire also included final-year students, the annual examinee figure was used as the most practical available estimate of the core target population for the application.

The sample size was considered using Cochran's formula, which is commonly used for estimating representative survey samples (Cochran, 1977). Using the conventional Cochran parameters of a 95 percent confidence level, a 5 percent margin of error, and an assumed population proportion of 0.5, the initial recommended sample size for a large population is approximately 384 respondents. Since the estimated annual Exit Exam population is finite, the finite population correction was then applied using $N = 7000$, resulting in an adjusted recommended sample size of approximately 364 respondents. This number represents the ideal sample size for a more statistically representative survey of Exit Exam candidates.

However, due to practical constraints such as limited time, limited access to respondents, and the limited project timeline, the study collected 47 valid responses. Data collection was focused on respondents who matched the target profile of medical graduates and final-year students. As a result, the sample

should be understood as a convenience sample rather than a fully representative national sample.

Despite this limitation, the collected responses were considered appropriate for the purpose of this project. The questionnaire was used as a requirements elicitation and preliminary user evaluation instrument, not as a nationwide statistical survey. The goal was to identify user needs, validate proposed features, and support early design decisions for Exit 360. Therefore, while the achieved sample size is smaller than the statistically recommended sample, it still provided useful evidence for understanding the needs and priorities of the target users. The limited sample size and convenience sampling approach are acknowledged as limitations of the research.

The Cochran sample size formula used for this estimate is shown below:

$$n_0 = \frac{Z^2 pq}{e^2}$$

where n_0 is the estimated sample size for a large population, Z is the z-score corresponding to the selected confidence level, p is the estimated proportion of the population with the attribute of interest, $q = 1 - p$, and e is the accepted margin of error. Since no prior population proportion was available, the default value $p = 0.5$ was used because it gives the maximum required sample size. For a 95 percent confidence level, $Z = 1.96$, $p = 0.5$, $q = 0.5$, and $e = 0.05$.

$$n_0 = \frac{(1.96)^2(0.5)(0.5)}{(0.05)^2} = \frac{3.8416 \times 0.25}{0.0025} = \frac{0.9604}{0.0025} = 384.16$$

Therefore, the recommended sample size is approximately 384 respondents. However, the actual sample size can be adjusted for the finite annual Exit Exam population using the following formula:

$$n = \frac{n_0}{1 + \frac{n_0 - 1}{N}}$$

where n is the adjusted sample size and N is the estimated population size. Using $N = 7000$:

$$n = \frac{384.16}{1 + \frac{384.16 - 1}{7000}} = \frac{384.16}{1.0547} \approx 364.2$$

Therefore, the adjusted recommended sample size is approximately 364 respondents. However, the actual number of valid responses collected was 47, which is why the survey findings are used for requirements elicitation and preliminary user evaluation rather than for making nationwide statistical claims.

3.1.3 Data Collection

Data for the user requirements part of the study were collected through a structured questionnaire. The questionnaire was selected because it allowed us to collect comparable responses from a group of respondents who were directly relevant to Exit Exam preparation within the limited project timeline. It also matched the purpose of this part of the research, which was to understand candidates' preparation conditions, their access to practice tests and internet, and their level of interest in the proposed features of Exit 360.

Two versions of the questionnaire were prepared. The English version was prepared for thesis documentation, while the Persian version was prepared for actual respondents so that the questions would be easier to understand and answer. Data collection was carried out through printed Persian copies distributed in person to eligible respondents. This approach was selected because the target respondents were more reachable through direct distribution, and it reduced the need for respondents to depend on an internet connection while completing the survey.

The questionnaire was titled Exit 360 Survey Questionnaire. It began with a short introduction explaining that the survey was part of a Kabul University Software Engineering capstone project for developing an Exit Exam preparation mobile application. The introduction also informed respondents that participation was voluntary, that the survey was anonymous, and that no personal identifying information would be published.

The questionnaire contained 28 items organized into five sections. The first section, Demographics, collected basic background information about the respondents. It asked about age, gender, original province, current study level, whether the respondent had already taken the Afghan Medical Exit Exam, and whether the respondent was attending an Exit Exam preparation course or coaching program. These items helped describe the respondent group and provided context for interpreting their answers.

The second section, Current Practice and Access, focused on the respondents' existing preparation habits and access conditions. It included questions about how often respondents take mock or practice tests, how easy it is for them to

access practice tests on demand, how they currently prepare for the Exit Exam, how often they have internet access for studying, how reliable their internet connection is, how important offline access is to them, and whether they have used any Exit Exam or medical exam preparation app before. This section was included because Exit 360 was intended to respond to real preparation barriers, especially access to practice questions, internet limitations, and the usefulness of offline support.

The third section, Feature Interest and Priorities, asked respondents to evaluate the planned features of Exit 360. It presented six proposed features: offline practice tests, performance analytics dashboard, subject-wise weak-area tracking, user profiles and peer following, online weekly competitive tests and leaderboards, and peer-to-peer challenge mode. Respondents rated the usefulness of each feature on a five-point scale from Not at all useful to Extremely useful. They were also asked to select and rank their top three most important features, and to indicate how likely they would be to use an Exit Exam preparation mobile application with these features. This section directly supported requirements prioritization because it showed which proposed features were most valued by potential users.

The fourth section, Perceived Impact, measured respondents' agreement with statements about the expected benefits of the application. The statements asked whether the app would improve exam preparation, help students practice more frequently, help identify strengths and weaknesses, help them focus on subjects where they need more practice, motivate more consistent study, increase exam confidence, and make it easier to study anywhere and anytime. These items used a five-point agreement scale from Strongly Disagree to Strongly Agree. This section helped capture the perceived educational value of the proposed system, not only interest in individual features.

The fifth section, Experience and Open Feedback, included open-ended questions. These questions asked respondents about the biggest barriers they face in preparing for the Exit Exam, what they liked and disliked about any Exit Exam or medical exam preparation apps they had used, what features would make them choose an exam preparation app over paper-based or center-based methods, and any additional comments or suggestions regarding an exam preparation platform. These open questions were included to allow respondents to express issues or ideas that might not be fully captured by fixed-choice questions.

Overall, the questionnaire combined multiple item types, including short-answer demographic items, multiple-choice questions, five-point usefulness

ratings, feature ranking, five-point agreement ratings, and open-ended questions. This structure allowed the questionnaire to collect measurable responses and additional comments while remaining short enough for students to complete within a reasonable time.

3.1.4 Data Analysis and Presentation

The closed-ended questionnaire responses were analyzed using descriptive statistics. Since the purpose of the questionnaire was exploratory requirements elicitation, the analysis focused on summarizing the responses clearly rather than making nationwide statistical claims. Categorical questions, such as study level, Exit Exam experience, preparation method, access to practice tests, internet access, offline-access importance, and prior use of Exit Exam or medical exam preparation apps, were summarized using frequencies and percentages.

For the five-point rating items, the analysis used mean and median scores to show the overall level of support for each proposed feature and each perceived-impact statement. In addition, responses were grouped where useful. For example, Very useful and Extremely useful responses were combined to show strong feature interest, while Agree and Strongly Agree responses were combined to show positive perceived impact.

The top-three feature priority question was analyzed using a weighted scoring method. A first-rank feature was assigned three points, a second-rank feature was assigned two points, and a third-rank feature was assigned one point. This made it possible to compare not only how often features were mentioned, but also how highly respondents prioritized them.

The results were presented through tables and charts generated from the questionnaire dataset. The charts were produced using Python from the original CSV file so that the analysis could be regenerated if the data changed. The results were organized in Chapter 4 according to the main questionnaire sections, including respondent profile, current practice and access, feature interest and priorities, perceived impact, and summary findings. This structure helped connect the survey results directly to the requirements of Exit 360.

3.1.5 Functional Requirements

The functional requirements describe the main services that Exit 360 must provide to its users. They were derived from the problem statement, the questionnaire findings, the review of existing Exit Exam and medical exam preparation applications, and the planned system scope described in the project

documentation.

FR-001: User Account and Profile Management

The system shall allow students to create an account, sign in, and maintain a basic profile. The profile should include information such as name, province, study level, university or preparation background, username, and optional profile image details. This information is needed for personalization, social features, competitive tests, synchronization, and progress tracking.

FR-002: Offline Question Bank Access

The system shall provide access to a bundled Exit Exam question bank inside the mobile application. Students should be able to view and attempt questions without requiring an active internet connection. This requirement is important because many target users have limited or unreliable internet access, and the core preparation experience should remain useful offline.

FR-003: Subject-Wise Practice Tests

The system shall allow students to practice questions by selecting specific subjects. Subject-wise practice should help students focus on weak areas, repeat difficult topics, and prepare across the clinical and paraclinical subjects included in the Exit Exam in a more organized way.

FR-004: General Mixed Practice Tests

The system shall allow students to take general mixed-subject practice tests that resemble the structure of the Exit Exam more closely than single-subject practice. Students should be able to configure basic test options such as the number of questions and whether a timer is used.

FR-005: Timed Test Sessions

The system shall support timed test sessions for both offline practice and online competitive tests. The timer should help students practice under exam-like pressure and should record the duration of each attempt so that results can be reviewed later.

FR-006: Automatic Grading and Result Display

The system shall automatically grade completed practice tests and display the result to the student. The result should include the total score, number of correct and incorrect answers, and where possible, subject-level performance. For Exit-style tests, the result should also support percentage-based interpretation so students can understand their performance in relation to the passing score.

For competitive online tests, trusted grading should be handled by the backend rather than the mobile client.

FR-007: Attempt History and Performance Analytics

The system shall store students' test attempts and present basic performance analytics. These analytics should include recent scores, score trends, average performance, stronger subjects, weaker subjects, and frequently missed questions. This requirement supports the need for faster feedback and better progress awareness during preparation.

FR-008: Offline Attempt Synchronization

The system shall allow attempts created offline to be stored locally and synchronized with the cloud account when internet access becomes available. Synchronization should avoid losing students' practice records and should keep local progress and cloud data reasonably consistent.

FR-009: Weekly Competitive Tests and Leaderboards

The system shall provide online weekly tests generated and controlled by the backend. Students should be able to join available weekly tests, submit answers, and view published leaderboard results after grading is complete. Leaderboards should be based on official backend results so that rankings remain fair and auditable.

FR-010: Peer Following and Challenge Tests

The system shall allow students to find and follow peers or friends, and to participate in friends-only challenge tests. Challenge tests should support asynchronous participation, so both students do not need to be online at the same time. This requirement supports motivation, comparison, and peer-based learning without making competition mandatory for all users.

FR-011: Exit Exam Blueprint Alignment

The system shall organize practice content according to the subject structure of the Afghan Medical Exit Exam. Questions should be grouped in a way that supports curative medicine and stomatology candidates, while also making the distinction between clinical and paraclinical subjects clear. This requirement helps ensure that practice inside the app remains connected to the actual exam structure rather than becoming a general medical quiz bank.

FR-012: Admin Management of Online Workflows

The system shall include an internal admin panel for controlled operational

tasks. Admin users should be able to inspect online data, manage weekly-test plans, trigger backend-owned question selection and leaderboard publishing actions, and review platform activity. This requirement supports the online and competitive parts of the system without exposing trusted operations to ordinary student accounts.

3.1.6 Non-Functional Requirements

The non-functional requirements describe qualities that the system should maintain while providing the features listed above. These requirements are written realistically for a student capstone project and avoid claims that would require large-scale production validation.

NFR-001: Offline Availability

The mobile application should allow students to use the core preparation features without an internet connection. Subject-wise practice, general practice, local grading, attempt history, and basic statistics should remain available offline after the question database has been bundled with the application.

NFR-002: Usability

The application should be simple enough for Exit Exam candidates to use without technical training. Common actions such as starting a practice test, selecting answers, viewing results, and checking progress should require only a small number of clear steps. The interface should use readable text, familiar controls, and consistent navigation.

NFR-003: Performance

The application should open local practice screens, load bundled questions, move between questions, and show results without noticeable delay on a typical Android phone. Online actions such as weekly-test submission, challenge updates, synchronization, and leaderboard loading may depend on internet quality, so the app should show clear loading and failure states when the network is slow.

NFR-004: Data Integrity

The system should preserve test attempts, selected answers, scores, and synchronization status accurately. It should reduce the risk of duplicate submissions, missing local attempts, or incorrect score records, especially when a user moves between offline and online use.

NFR-005: Security and Access Control

The system should protect student accounts and trusted online workflows. Ordinary users should not be able to read private answer keys, edit official scores, publish leaderboards, manage weekly tests, or perform admin actions. Competitive grading, leaderboard publishing, and challenge state changes should be handled through authorized backend operations.

NFR-006: Privacy

The system should collect only the student information needed for profile use, progress tracking, competition, and synchronization. Personal data should not be shown publicly unless it is part of an intended social or leaderboard feature.

NFR-007: Reliability

The application should handle common failure cases gracefully, including weak internet access, interrupted synchronization, expired challenge requests, and unavailable backend services. In such cases, the user should receive understandable feedback and should not lose locally saved practice data.

NFR-008: Maintainability

The system should be organized so that the mobile app, backend services, admin panel, and local data layer can be developed and tested separately. This supports the iteration-based development approach and makes it easier to update the question bank, Exit Exam subject structure, competitive workflows, and administrative features over time.

3.1.7 Data Requirements

Exit 360 depends on several types of data. Some of this data is needed for offline practice, some is needed for online competition and synchronization, and some is needed for managing the Exit Exam subject structure. The main data requirements are described here at a requirements level, while the detailed storage design is discussed later in the database design section.

DR-001: Question Bank Data

The most important educational data required by the system is the Exit Exam question bank. Each question should include the question text, answer options, correct answer, subject, topic where possible, difficulty level where available, and any explanation or metadata needed for review. The offline version of the application requires a bundled local question bank so that students can take

practice tests without internet access.

At the current stage, the available question bank is limited. Building a complete question bank for the Afghan Medical Exit Exam is challenging because the exam covers a wide range of clinical and paraclinical subjects. It would not be appropriate to rely on leaked exam forms or copied online book material without clear permission, since that would raise ethical and copyright concerns. A larger and more reliable question bank would require expert support from medical teachers, subject specialists, or authorized educational sources to ensure accuracy, fairness, and proper coverage of the exam blueprint.

DR-002: Subject and Exam Blueprint Data

The system requires structured subject data so that questions can be grouped and filtered correctly. This includes subjects, topic labels, clinical or paraclinical grouping, and any available mapping to the Exit Exam subject breakdown. This data supports subject-wise practice, mixed tests, analytics by subject, weak-area tracking, and future expansion of the question bank.

DR-003: User Profile Data

The system requires basic profile data for students who use online features. This may include name, username, province, university or preparation background, study level, graduation year, and optional profile image information. Profile data is needed for account personalization, peer features, leaderboards, challenges, synchronization, and progress continuity across devices.

DR-004: Practice Attempt and Performance Data

The application needs to store students' practice attempts, selected answers, scores, duration, test type, subject coverage, and attempt date. This data is required for result display, attempt history, progress tracking, performance analytics, and identifying weak subjects or frequently missed questions. Offline attempts should be stored locally first and then synchronized when the student has internet access.

DR-005: Online Competition Data

The online layer requires data for weekly tests, test sessions, submissions, leaderboard snapshots, and peer challenge records. Weekly tests need information such as title, schedule, duration, question count, subject coverage, and publication status. Test sessions and submissions must store official participation and grading records so that competitive results can be trusted.

DR-006: Question Management and Review Data

The system requires data that helps maintain and review the question bank over time. This may include question source notes, subject tags, review status, correction history, and optional explanation text. This data is important because medical exam questions must be accurate, properly categorized, and aligned with the intended Exit Exam subject area before they are used for practice or online competition.

DR-007: Admin and Operational Data

The system requires internal administrative data for managing weekly tests, inspecting question summaries, publishing leaderboards, reviewing submissions, and supporting controlled backend workflows. This data should only be accessible to authorized admin users because it may affect trusted test content and official competitive results.

DR-008: Data Sensitivity and Protection

Different types of data require different levels of protection. Public or low-risk data includes general subject names and basic published leaderboard information. Sensitive data includes user profiles, scores, challenge records, and synchronization records. Highly restricted data includes correct answer keys, private question bank content for competitive tests, admin operations, official submissions, and leaderboard publishing actions. The system should therefore separate local practice data from trusted online data and should prevent ordinary users from accessing answer keys or editing official results.

3.2 Development Methodology

The development of Exit 360 followed an Agile development methodology with a lightweight Scrum-based organization. The project was not treated as one long fixed sequence of tasks. Instead, the work was divided into manageable iterations so that the offline mobile foundation, online services, and admin workflows could be designed, implemented, tested, and improved step by step. This approach was suitable for a student capstone project because the team was small, the available time was limited, and the requirements became clearer as the questionnaire, literature review, prototype work, and supervisor feedback progressed.

3.2.1 Agile Development Approach

Agile development was suitable for Exit 360 because the system combines several connected but separable parts, including offline question practice, local storage,

progress statistics, online competition, synchronization, and administrative workflows. These parts depend on a shared application structure, but they do not all need to be completed at the same time. Therefore, the project began with the most important user value, which was offline Exit Exam practice, and then expanded toward online and administrative features.

This approach also matched the requirements elicitation findings. The questionnaire focused on preparation habits, access to practice tests, internet reliability, offline access, and interest in features such as analytics, weekly tests, leaderboards, and challenge tests. By working iteratively, these findings could be translated into requirements and then into implemented features in a practical order. This reduced the risk of spending too much time on secondary features before the core preparation workflow was working.

3.2.2 Scrum-Based Project Organization

The project used Scrum concepts in a lightweight way. As described by Radaideh (2024), Scrum work is commonly organized around a project vision, prioritized product backlog, sprint planning, sprint backlog, deliverable creation, review, and retrospective improvement. Exit 360 followed this structure at a student project scale rather than as a full corporate Scrum process.

The product vision was defined as an offline-capable Exit Exam preparation mobile application with optional online competition. The product backlog contained the main functional requirements, non-functional requirements, data requirements, user interface tasks, backend tasks, admin tasks, and testing tasks. Before each development cycle, the team selected a smaller set of related tasks and treated them as the sprint backlog for that iteration.

The Scrum principles were applied practically. Transparency was maintained through shared project documents, Git history, and regular discussion of current progress. Inspection was performed through prototype review, testing, supervisor feedback, and checking whether the implemented features matched the requirements. Adaptation happened when requirements or design choices were adjusted after review, such as prioritizing offline preparation and refining the online competition layer. Collaboration was important because the project involved a Flutter mobile app, Firebase backend services, a React web admin panel, questionnaire work, and LaTeX thesis documentation that had to remain consistent with each other. Value-based prioritization was reflected in implementing the offline preparation workflow first, because the questionnaire and project scope showed that offline practice was the most important user need. Time-boxing was reflected in dividing the work into limited development

iterations of approximately one to one and a half months, even though the project did not follow a strict corporate sprint schedule.

Because the project team consisted of three students, Scrum roles were handled in a simplified way. The team did not maintain separate full-time corporate roles. Instead, product-owner-like responsibilities were reflected in keeping the Exit 360 scope and requirements clear, Scrum-master-like responsibilities were reflected in coordinating tasks and removing blockers, and development responsibilities were shared across implementation, testing, and documentation. This kept the method realistic for a capstone project while still following the main Scrum ideas of prioritization, time-boxing, review, and iterative improvement.

3.2.3 Development Iterations

The development work was planned in three main iterations. Although the iteration lengths were not identical, each major iteration lasted approximately one to one and a half months because the project combined implementation, testing, documentation, and user review within the same academic timeline. The first iteration focused on the offline foundation of the Flutter mobile application, including navigation, local Exit Exam question storage, subject-wise practice, mixed tests, timers, local grading, result summaries, answer review, attempt history, and basic statistics. This iteration was important because it made the application useful even without internet access.

The second iteration focused on online and social features. This included Firebase setup, authentication, user profiles, peer workflows, backend-generated weekly tests, server-side grading, leaderboard snapshots, asynchronous challenges, notifications, synchronization, and Firebase security validation.

The third iteration focused on the internal admin panel and operational workflows. The admin panel was planned to support controlled management of weekly tests, question summaries, leaderboard publishing, users, sessions, submissions, and challenges. This reduced the need for manual database edits and helped keep trusted online workflows under backend control.

3.2.4 Prototype Development and User Review

The iterative development process was also reflected in the prototypes prepared during the project. Each major development iteration produced a prototype that represented the main focus of that stage. The first prototype focused on the offline preparation experience, the second prototype extended the system with online and social features, and the third prototype connected the mobile

application with the admin and operational workflows. This made it possible to review the project in smaller working parts rather than waiting until the whole system was complete.

Each prototype was reviewed through a focus group with six participants. During these reviews, the prototype purpose and main features were presented, participants discussed their feedback, and the team identified the changes needed before the next iteration. This helped connect each development stage with practical evaluation and improvement before the next iteration began.

3.2.4.1 Prototype 1: Offline Preparation Experience

The purpose of the first prototype was to validate the core offline preparation workflow of Exit 360. This prototype focused on the part of the application that should remain useful even when the user has no internet connection. It included local question access, subject-wise practice, mixed practice test creation, test configuration, timed test taking, automatic grading, result display, answer review, attempt history, and basic performance statistics.

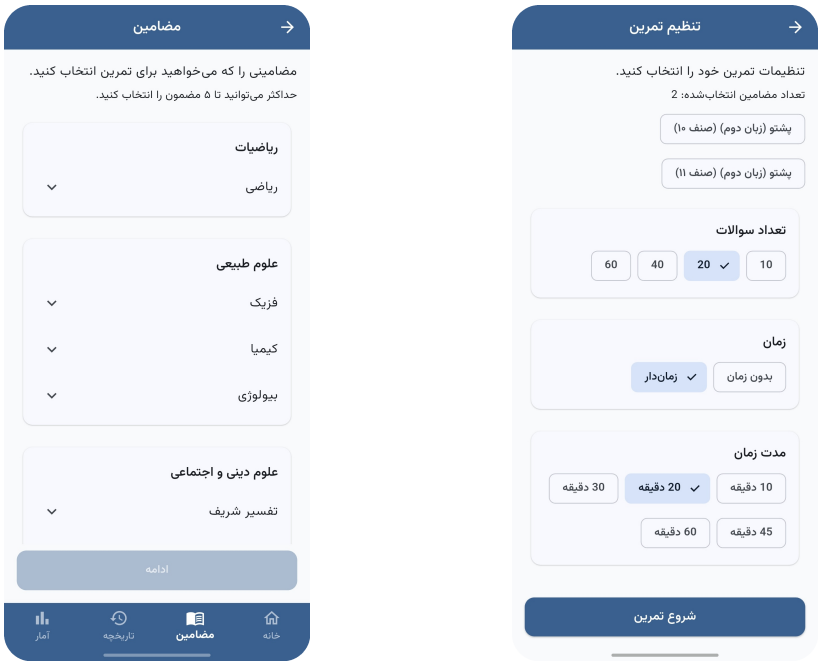
The first prototype was generally well received because it showed that an Exit Exam candidate could open the app, select practice content, complete a test, view the result, and review answers without depending on Firebase or a live internet connection. This feedback supported the offline-first direction of the project, since offline practice had already been identified as one of the most important needs for the target users.

At the same time, user review identified several improvement points. The available question bank was very limited and was included mainly for testing the prototype workflow. It was also available only in Dari, which limited how fully the offline preparation experience could be evaluated for a broader group of users. The subject selection screen also needed refinement because the subjects were considered nested too deeply across separate screens. Feedback also suggested improving the results, history and statistics screens so that the information would be easier to scan and understand. In the review flow, users requested filters for correct, incorrect, and unanswered questions, as well as deleting unwanted history records and an incorrect-only test option that would allow them to practice again using only previously missed questions instead of just reviewing those questions.

The feedback from the first prototype led to refinements in the planned offline workflow. The offline-first direction was kept as the foundation of the system, and more Dari questions were added in alignment with the Exit Exam subject

structure. However, building a comprehensive bilingual question bank in Dari and Pashto was left as future work because it required expert medical assistance and careful content validation. Some Exit Exam preparation books and question collections were available, but direct question mining from those sources was avoided for ethical and copyright reasons. Moreover, subject organization; results, history and statistics presentation; review filtering, history attempt removal; and missed-question practice were also treated as important improvements for later development. The screenshots below show the first prototype in more detail.

The first part of the main practice flow is shown in Figure 3.1. These screens show how the user selected practice content and configured a practice test before starting the test session.



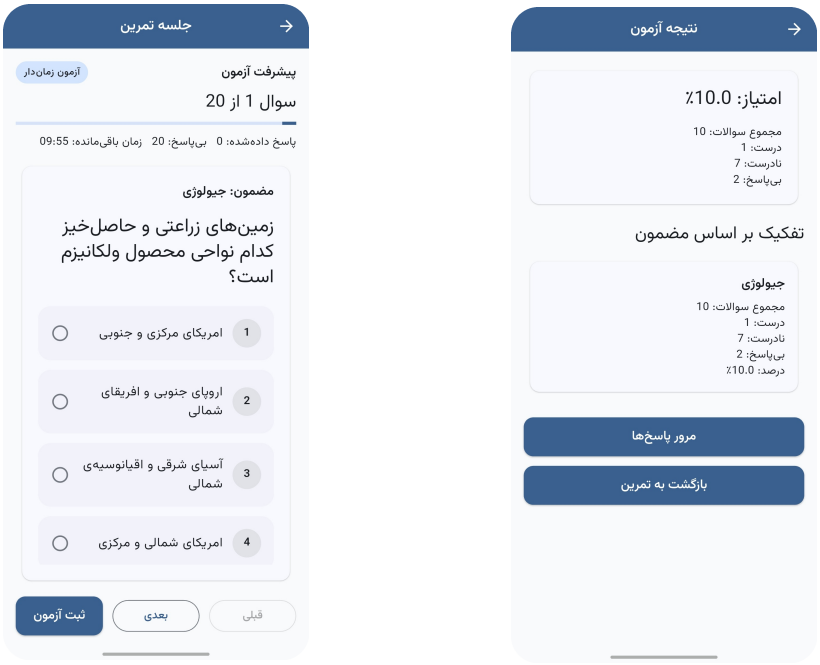
Subject Selection Test Configuration

Figure 3.1: Prototype 1 practice subject selection and test configuration screens

As shown in Figure 3.1, the subject selection screen first separated practice content into curative medicine and stomatology paths, then moved users into separate clinical and paraclinical subject screens before they could select the actual subjects. Users could select up to three subjects at once. This structure supported organized practice, but it also showed why users felt that subject navigation was nested too deeply and needed to be simplified.

The test configuration screen in Figure 3.1 displayed the selected subjects and allowed users to choose the number of questions, including 10, 20, 40, or 60 questions. It also allowed the test to be taken with or without a timer. When timed mode was selected, the available duration options were 10, 20, 30, 45, or 60 minutes. Together, these two screens demonstrated the main configuration path of the first offline prototype while also revealing the need for a more direct subject selection experience.

The next part of the Prototype 1 flow is shown in Figure 3.2. These screens show the actual test-taking interface and the result screen presented after submission.



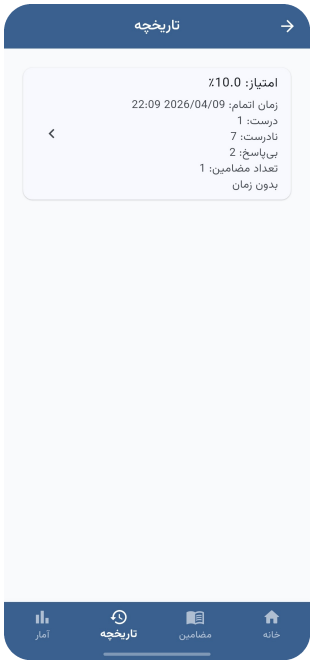
Test Taking Test Result

Figure 3.2: Prototype 1 practice test taking and test result screens

As shown in Figure 3.2, the test-taking screen presented the main controls needed during an offline practice session. It showed the test progress bar, the number of answered and unanswered questions, and, when timed mode was enabled, a timer label and remaining-time counter at the top of the screen. In timed mode, the test was also submitted automatically when the time expired. The question card displayed the subject name, the question text, and four answer choices. The screen also provided previous, next, and submit buttons. Users were allowed to submit at any time, so the offline practice did not force them to answer all questions before finishing the test.

The test result screen in Figure 3.2 showed the final percentage, the total numbers of correct, incorrect, and unanswered questions, and a subject-wise breakdown of performance. It also provided navigation paths to review the completed attempt or return to the practice home screen. The result interface used simple cards rather than graphs, colorful badges, or complex visual summaries. This kept the first prototype easy to understand, while also showing that the result and feedback screens could be improved in later iterations.

The attempt review part of Prototype 1 is shown in Figure 3.3. These screens show how completed attempts were listed and how a selected attempt could be opened for answer review.



Practice History



Practice Review

Figure 3.3: Prototype 1 practice history and review screens

As shown in Figure 3.3, the practice history screen presented previous test attempts using simple cards. Each card showed the result percentage and the total numbers of correct, incorrect, and unanswered questions. It also indicated whether the attempt was timed or untimed, displayed the total time for timed attempts, and included the completion timestamp. Tapping a history card opened the review screen for that specific attempt. However, the history screen did not yet support swipe removal for unwanted attempts, which was another small improvement requested during review.

The review screen in Figure 3.3 showed a progress bar and a question card containing the subject name, question text, and answer status. The status indicated whether the question was correct, incorrect, or unanswered. For correct and unanswered items, the screen showed the correct answer. For incorrect items, it showed both the selected incorrect answer and the correct answer, with an explanation when one was available. The screen also provided previous, next, and back buttons for navigation. However, the first prototype did not include filters for viewing only correct, incorrect, or unanswered questions, which became one of the requested improvements after user review.

The statistics screen of Prototype 1 is shown in Figure 3.4. This screen presented the first version of performance summaries for offline practice attempts.



Figure 3.4: Prototype 1 practice statistics screen

As shown in Figure 3.4, the statistics screen also used simple text and number cards rather than graphical elements. It displayed the total number of tests, average score percentage, best score percentage, and last attempt percentage. It also showed the score trend, indicating whether performance was increasing,

stable, or decreasing. In addition, it identified the strongest subject and weakest subject with their average percentages. This screen provided an early form of progress awareness, but its simple presentation also supported the feedback that progress information could be made easier to scan in later iterations.

3.2.4.2 Prototype 2: Online Social and Competition Features

The purpose of the second prototype was to validate the online and social layer of Exit 360. This prototype extended the offline preparation foundation by adding account-based access, Firebase integration, peer interaction, weekly tests, leaderboards, friend requests, and challenge-test workflows. At the time of the review, the application was in the middle of a design and theme overhaul, so some interface screens were still not visually polished and general user interface improvements were expected. Therefore, the review did not focus only on the visible mobile interface. It also considered backend validation because the online competition features depended heavily on trusted server-side behavior. This included checking whether Firebase, Cloud Functions, and Firestore Rules correctly protected online sessions, submissions, challenge states, leaderboard data, and other backend-controlled workflows from unsafe client-side access.

The authentication flow was still at a smoke-test stage and the focus group feedback confirmed it. Since access to the online competition section required an account, users first interacted with the authentication screens. These screens were sufficient for verifying that sign-in, sign-up, and sign-out worked, but they were not yet complete as a user-ready experience. Localized error messages, email verification, forgot-password support, and profile completion fields such as full name, username, province, and related user details still needed to be integrated.

The online layers for friends, challenges, weekly tests, and leaderboards worked functionally during the prototype review. However, the review also identified interface, rules, and notification issues that needed refinement. The competition home screen received mixed reactions. Users liked that it placed the main online features in one location using action cards for weekly tests, leaderboards, friend requests, and challenges. At the same time, the number of Firestore reads needed to load several competition summaries made the screen relatively slower to open. The review also identified a navigation inconsistency. The practice section used a centralized bottom navigation structure in addition to links from the practice home screen, while the competition section was organized as one long screen where cards were the only path to dedicated online screens. This showed that the online section needed a more consistent navigation pattern.

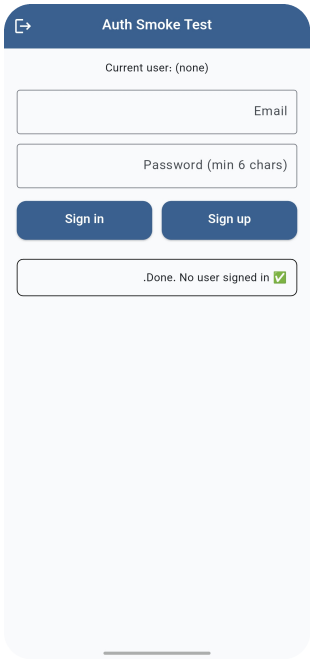
The rules issue appeared most clearly through data-display problems in the social and challenge workflows. Searching for friends by username returned readable usernames and full names, but incoming requests, outgoing requests, and existing friend cards showed unreadable user identifiers because of a Firestore Rules and data-access mismatch. The same problem appeared in the challenge workflow when creating a challenge, viewing received challenges, and showing the challenge result. In the result screen, displaying the winner by user identifier made it unclear who had actually won the challenge. The review also pointed out the absence of notifications for events such as a weekly test going live, leaderboard publication, challenge acceptance, and challenge completion, since the planned Firebase Cloud Messaging (FCM) support had not yet been integrated.

The feedback from the second prototype led to several planned refinements. The authentication flow needed to be completed with localized errors, email verification, password recovery, and profile completion. The competition home needed optimization to reduce unnecessary Firestore reads or load summaries more efficiently. The navigation structure of the competition section needed to be made more consistent with the rest of the mobile application. Notification support also needed to be added for important online events. Finally, the social and challenge workflows needed corrected Firestore access rules and data models so that usernames and full names could be shown where users needed to identify friends, opponents, request senders, and challenge winners.

The first two screens of the online prototype are shown in Figure 3.5. These screens show how users entered the online layer through authentication and then reached the central competition home screen.

As shown in Figure 3.5, the authentication screen was mainly a smoke-test interface for Firebase Authentication. It allowed users to create an account with an email and password, sign in, and sign out. It also checked basic cases such as email format, password policy, and previously used emails, but these messages were still displayed through Firebase's native error responses. After successful sign-in, the screen could print the user identifier for verification. At this stage, the flow did not yet include email verification, forgot-password support, profile completion, full name entry, username claiming, province, university, or graduation-year details.

The competition home screen in Figure 3.5 acted as the main entry point for the online layer. Its app bar included refresh, profile, and entry-home actions. Below the app bar, a decorative card briefly described the screen and displayed the user's full name. The rest of the screen was organized as a set of summary



Authentication

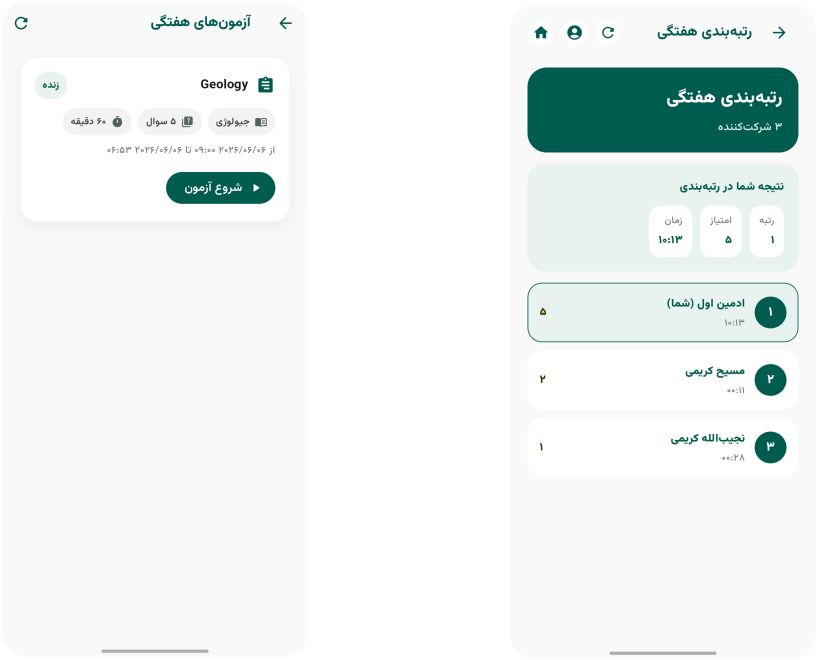


Competition Home

Figure 3.5: Prototype 2 authentication and competition home screens

cards that gave users a quick overview of the main online features. A status card showed badges for login state, last synchronization timestamp, and total subjects. The weekly-test card showed the latest available test with its subject, question count, duration, start time, end time, and status such as scheduled, live, or leaderboard released. The latest leaderboard card showed the total number of weekly test participants and the names and scores of the top ranked users. The friends card displayed total friends and total requests and also included a username search field directly inside the card. Finally, the challenges card summarized pending, active, and completed challenges and listed challenge rows with status, total questions, and time. Each summary card also provided buttons that took users to the related dedicated screen, but this was the only navigation path because the competition section did not yet have a centralized bottom navigation structure. This all-in-one design made the online features visible from one place, but it also contributed to the slower loading behavior noted during review because several Firestore summaries were loaded together.

The weekly test and leaderboard screens of Prototype 2 are shown in Figure 3.6. These screens show how users accessed an available weekly test and viewed ranked results after leaderboard publication.



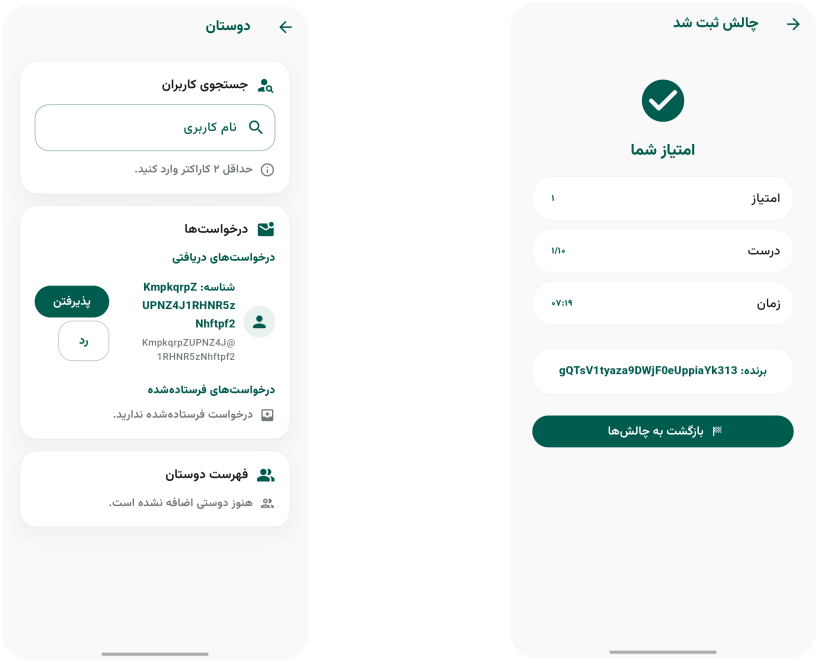
Weekly Tests Leaderboards

Figure 3.6: Prototype 2 weekly test and leaderboard screens

As shown in Figure 3.6, the weekly test screen included a refresh action in the app bar and displayed available weekly tests as cards. Each card showed the subject name, question count, duration, start time, end time, and current status, similar to the weekly-test summary on the competition home screen. The dedicated weekly test screen also added a start test button, allowing users to begin an available online test from the weekly-test list.

The leaderboard screen in Figure 3.6 showed the total participants for the selected weekly test and then displayed the full leaderboard. Each leaderboard row included participant details such as name, total time taken, and score. The ranking was sorted by higher score first, with quicker submission time used as the tiebreaker when participants had the same score. Before the full leaderboard, a summary card identified the current user. When the user had participated, this card showed the user's rank, score, and time, and the same user's row was highlighted inside the full leaderboard. When the user had not participated, the screen showed a message instead of personal ranking details.

The friends and challenge result screens of Prototype 2 are shown in Figure 3.7. These screens are related because a user needed to have a friend before creating a peer challenge.



Friends Challenge Result

Figure 3.7: Prototype 2 friends and challenge result screens

As shown in Figure 3.7, the friends screen included a username search field at the top. This search correctly returned relevant users with their usernames and full names. The requests card showed incoming requests with accept and decline actions, and outgoing requests with a cancel action. However, because of a Firestore Rules and data-access mismatch, the request cards showed Firebase user identifiers instead of readable usernames or full names. The same issue appeared in the friends card, where existing friends were also listed using unreadable identifiers.

The challenge result screen in Figure 3.7 represented the final step after a user created a challenge by selecting a friend as the opponent, choosing subjects, setting the question count, and selecting the time limit. The challenge would proceed only after the opponent accepted it. After the user completed the challenge, the result screen immediately showed the user's score, correct count out of the total questions, and time. If the opponent had also completed the challenge, the screen displayed the winner. However, because the same readable-profile issue affected the challenge workflow, the winner could only

appear as an unreadable user identifier. If the opponent had not completed the challenge yet, the screen showed a waiting-for-opponent message instead. This confirmed that the challenge workflow was functionally connected, but that profile lookup and display rules needed to be corrected before the feature could feel clear to users.

3.2.4.3 Prototype 3: Admin and Operational Workflows

The purpose of the third prototype was to validate the internal admin side of Exit 360 and its connection with the online competition workflows. This prototype focused on the controlled management of the cloud-based question bank, users, submissions, results, challenges, weekly tests, leaderboards, and other backend-owned operations that should not be handled directly from normal student accounts. The admin panel was created so that privileged work could be carried out through a controlled interface instead of directly editing Firestore records. Since this prototype was an internal operational tool rather than a student-facing feature, its focus group review was held with fellow developers instead of ordinary app users.

The admin panel was well received for providing read-only Firestore summaries about questions, users, submissions, results, challenges, and weekly tests through dedicated pages and a comprehensive dashboard. These summaries were useful for inspection, but the weekly-test workflow went beyond read-only viewing by supporting controlled admin actions. Admins could plan a weekly test by selecting the subject coverage, question count, duration, start time, and end time. After the test became live, the admin workflow could generate the question set, and after the test expired, it could publish the leaderboard. This confirmed that the admin panel could control the main weekly-test lifecycle without requiring direct manual edits in Firestore for every operation.

However, the weekly test workflow still had usability issues. The weekly-test creation screen required the admin to manually enter internal Firestore subject, class, and group identifiers. Although these identifiers were custom readable IDs rather than unreadable Firestore auto IDs, the process was still error-prone and difficult. The focus group suggested that the admin panel should provide a proper subject selection interface similar to the mobile app, so that internal IDs would be hidden from the admin user. The same issue appeared in the filtering controls. Although weekly tests could be filtered by status through a dropdown, the filter for subject, class, and group still expected manually written IDs. These filters were therefore suggested to be changed to readable dropdown selections.

Another major concern was the question bank page. The page provided useful

summaries based on subjects, classes, and groups, including total questions, total active questions, and badges showing whether the available questions were enough for creating a large weekly test. However, the page did not allow admins to upload or add new questions through the panel. This meant that question-bank expansion still required direct Firestore manipulation. While read-only summary pages were acceptable for many admin views, the question bank page needed to support adding new questions because question management is a core operational task for the system.

A final issue was admin account management. The prototype allowed login only for the sole admin account that had been manually created in Firestore through custom admin claims. It did not provide a way to create another admin account from the panel or change the password of the admin accout. This again made the system depend on direct Firestore or Firebase-side manipulation for an administrative task that should later be handled through a controlled workflow.

The feedback from the third prototype therefore led to several planned refinements. The admin panel needed a more user-friendly weekly-test planning interface, readable dropdown filters, question-bank expansion support, and a controlled way to manage additional admin accounts. These changes were important because the admin panel was intended to reduce direct Firestore editing, not simply move some operational tasks into the browser while leaving other important tasks manual.

The dashboard page of the admin prototype is shown in Figure 3.8. This page provided the main overview of the admin panel with a collapsible sidebar, currently expanded, offering navigation links to the other admin pages.

As shown in Figure 3.8, the admin panel used a collapsible sidebar with links to the main pages of the panel. The dashboard itself began with summary cards showing the total number of users, weekly tests taken, total questions in the bank, active questions, submitted results, and challenges between users. In this context, submitted results referred to online test sessions that had been securely graded on the server.

The lower part of the dashboard in Figure 3.8 provided additional operational summaries. A table showed the latest weekly tests with fields such as identifier, title, status, start time, question count, and whether the questions had been generated. A bar chart summarized weekly tests by status, such as scheduled, live, closed, and published, while a donut chart summarized challenges by status, including pending, active, completed, declined, canceled, and expired. The charts revealed exact values when hovering over a bar or segment, but

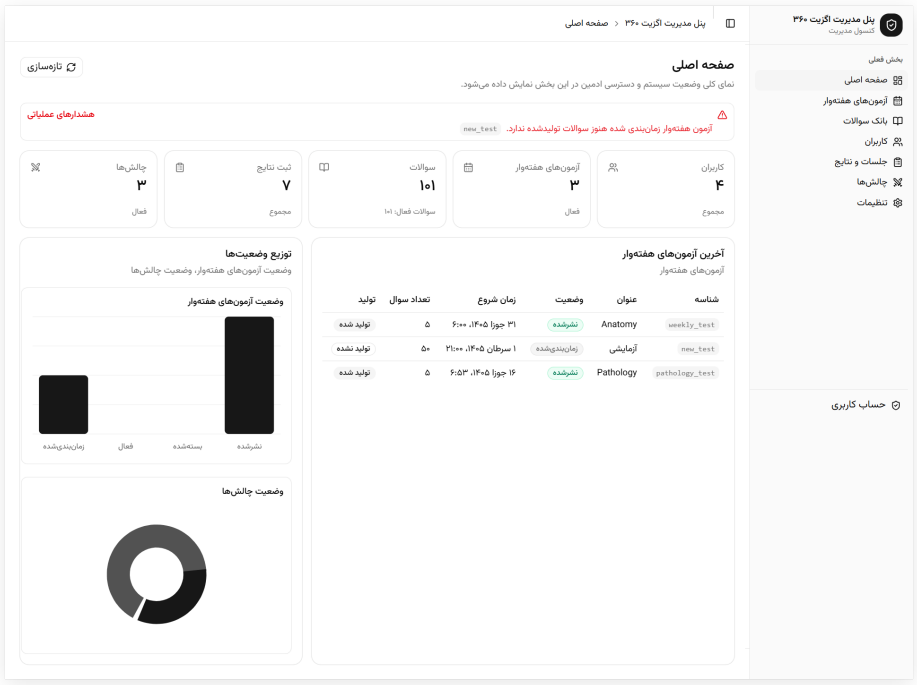


Figure 3.8: Prototype 3 admin dashboard page

the donut chart did not include a visible legend, so the meaning of each color was not immediately clear without interaction. The dashboard also showed operational alerts, such as scheduled weekly tests whose questions had not yet been generated and closed weekly tests whose leaderboards had not yet been published.

The weekly tests page of the admin prototype is shown in Figure 3.9. This page was important because it represented the main operational workflow for planning weekly tests and controlling their question generation and leaderboard publication.

As shown in Figure 3.9, the page provided a summary table for all weekly tests. The table included the weekly test identifier, title, status, class identifier, group identifier, subject identifier, start and end times, question count, and badges showing whether the question set had been generated and whether the leaderboard had been published. Each row also included an action menu under a triple-dot button. This menu supported actions such as viewing details, editing the test, generating the question set, and publishing the leaderboard. The last three actions were available only when the weekly test was in the scheduled state, which helped prevent actions from being performed at the wrong stage of the weekly-test lifecycle.

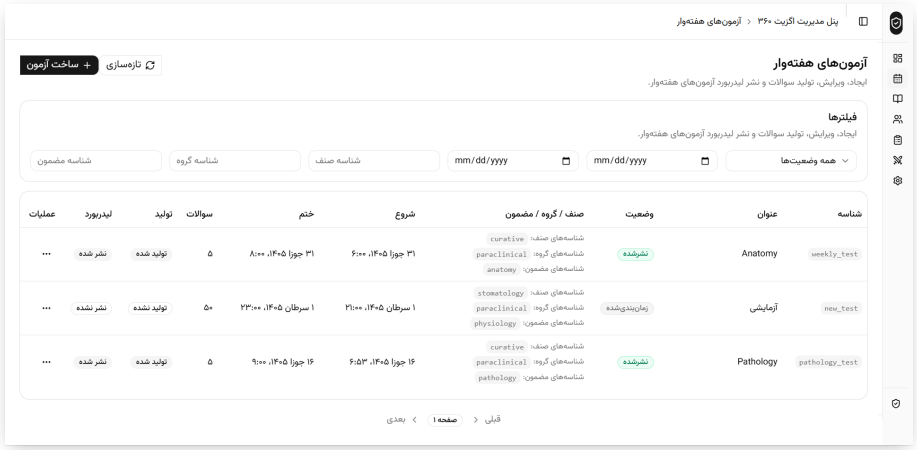


Figure 3.9: Prototype 3 admin weekly tests page

The top of the page in Figure 3.9 also included a refresh button and a button for creating a new weekly test. Above the table, the page provided filters for weekly test status, start and end times, class, group, and subject. The status filter was easier to use because it used a dropdown, but the class, group, and subject filters required manually written internal identifiers. Although the page was functionally useful for inspecting and controlling weekly tests, as brought up during the prototype review, the ID-based filters needed to be replaced with dropdown selections so that admins could work with familiar names rather than internal Firestore IDs. Even though the identifiers were custom readable values, the admin still had to know and type the exact values manually, which increased the chance of mistakes and slowed down filtering.

The weekly-test creation workflow is shown in Figure 3.10. This modal dialog appeared after the admin clicked the create weekly test button on the weekly tests page shown earlier in Figure 3.9.

As shown in Figure 3.10, the modal dialog allowed the admin to enter the weekly test identifier, title, question count, duration, subject identifier, class identifier, group identifier, start timestamp, and end timestamp. It also included toggles for shuffling the question order and the answer-choice order. The timestamp fields used Gregorian date and time pickers, while also showing the Jalali alternative directly below the selected Gregorian value. This made the scheduling workflow more understandable in the local context because admins could see both calendar representations while preparing a test.

However, the same identifier problem appeared in this modal dialog. The subject, class, and group fields required manual entry, and multiple values had

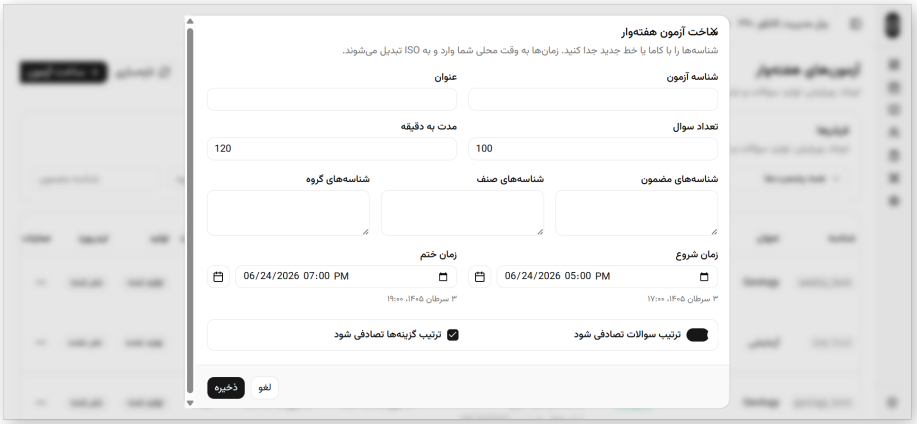


Figure 3.10: Prototype 3 weekly test creation modal dialog

to be typed as comma-separated values or on separate lines. During the review, this was identified as unnecessary friction for an admin-facing workflow. The suggested improvement was to replace these fields with readable dropdown selections or a separate subject selection dialog. A clearer flow would first allow the admin to choose the class, such as Curative Medicine or Stomatology, then choose the group, such as Paraclinical or Clinical, and finally select the subject from a narrowed-down list of eligible subjects. This would hide internal identifiers and make weekly-test creation easier and less error-prone.

The challenges page of the admin prototype is shown in Figure 3.11. This page represented one of the read-only summary views used to inspect online peer challenge activity across the system.

As shown in Figure 3.11, the challenges page summarized the challenge requests created between users. The main table included information such as the challenge identifier, challenger identifier, opponent identifier, status, subject identifier, question count, challenge creation time, acceptance deadline, completion deadline, result, and a button for viewing more details. This made the page useful for inspecting the state of challenge workflows without allowing admins to directly change user challenge records from the summary table.

The upper part of Figure 3.11 also included a refresh button and filters for narrowing the table. Admins could filter challenges by status, participant user identifiers, and a selected time range. This supported basic inspection of challenge activity, but it also showed the read-only nature of this part of the prototype. Unlike the weekly-test page, which included operational actions, the challenges page mainly helped admins observe whether challenge requests, deadlines, statuses, and results were being recorded correctly.

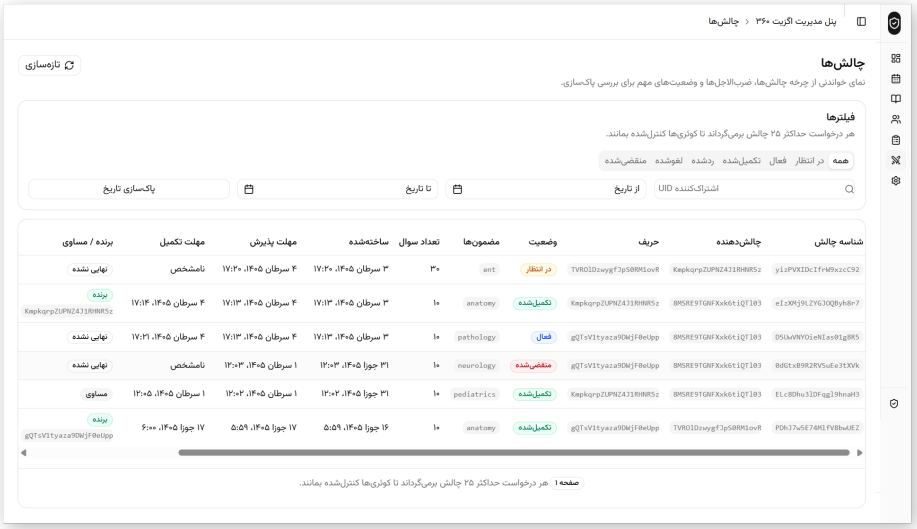


Figure 3.11: Prototype 3 admin challenges page

3.3 System Design

This section presents the main design models used to describe the structure and behavior of Exit 360 before implementation. The design explains how the main actors interact with the system, how the application components are organized, and how important workflows are expected to operate. These models provide a bridge between the requirements identified earlier and the technical implementation discussed in the later sections.

3.3.1 Use Case Model

The use case model identifies the main interactions between Exit 360 and its external actors. As shown in Figure 3.12, the system has two primary human actors: the student and the admin. The student actor represents Exit Exam candidates who use the mobile application for offline practice, online participation, peer interaction, and progress tracking. The admin actor represents authorized internal users who manage trusted operational workflows such as question-bank management, weekly-test planning, leaderboard publishing, and platform review.

As shown in Figure 3.12, the student-side use cases are grouped around two main areas. The first area is offline learning, where students take subject-wise and general practice tests, use timed sessions, view results, review attempt history, and check performance analytics. These use cases support the offline-first goal of the system because they allow students to continue preparing even when

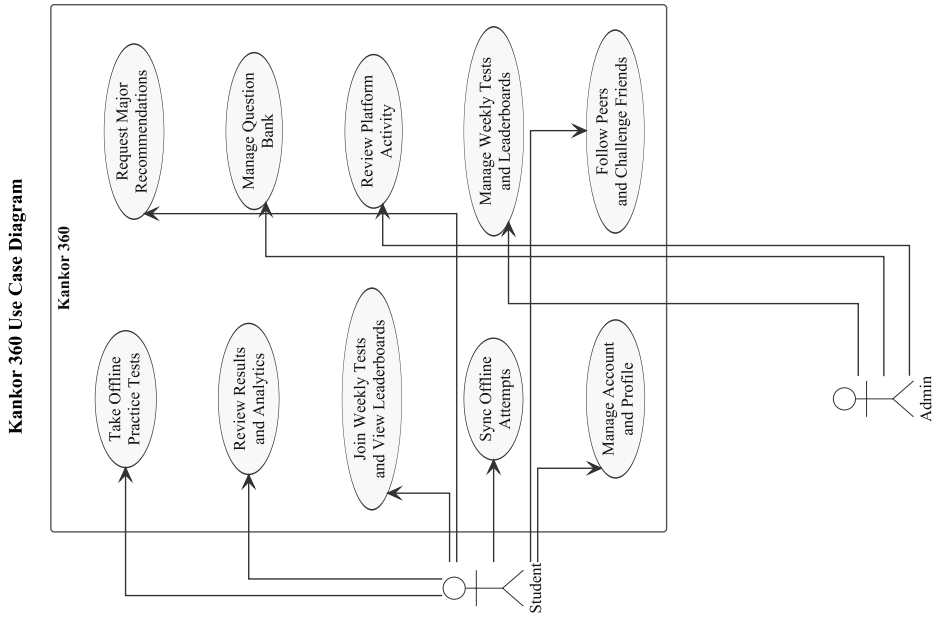


Figure 3.12: Use Case Diagram

internet access is unavailable.

The second area is the online and social layer. It includes registration, profile management, synchronization of offline attempts, weekly competitive tests, leaderboards, peer following, friend challenges, and notifications. In implementation, these use cases depend on backend services because competitive grading, leaderboard publication, challenge state, and synchronization require trusted online processing.

The admin-side use cases are separated from the student-facing features because they affect trusted system data. Admins manage the question bank, plan weekly tests, publish leaderboards, review users and submissions, and monitor platform statistics. Separating these operations in the use case model helps clarify that ordinary students should not have direct access to administrative or trusted competitive workflows.

3.3.2 Sequence Diagrams

Sequence diagrams were prepared for the main workflows that require a clear view of interaction between the user interface, application logic, storage, and backend services. Figure 3.13 presents the first sequence diagram, which focuses on the offline practice-test flow because this is the foundation of the application and the part that must remain usable without internet access.

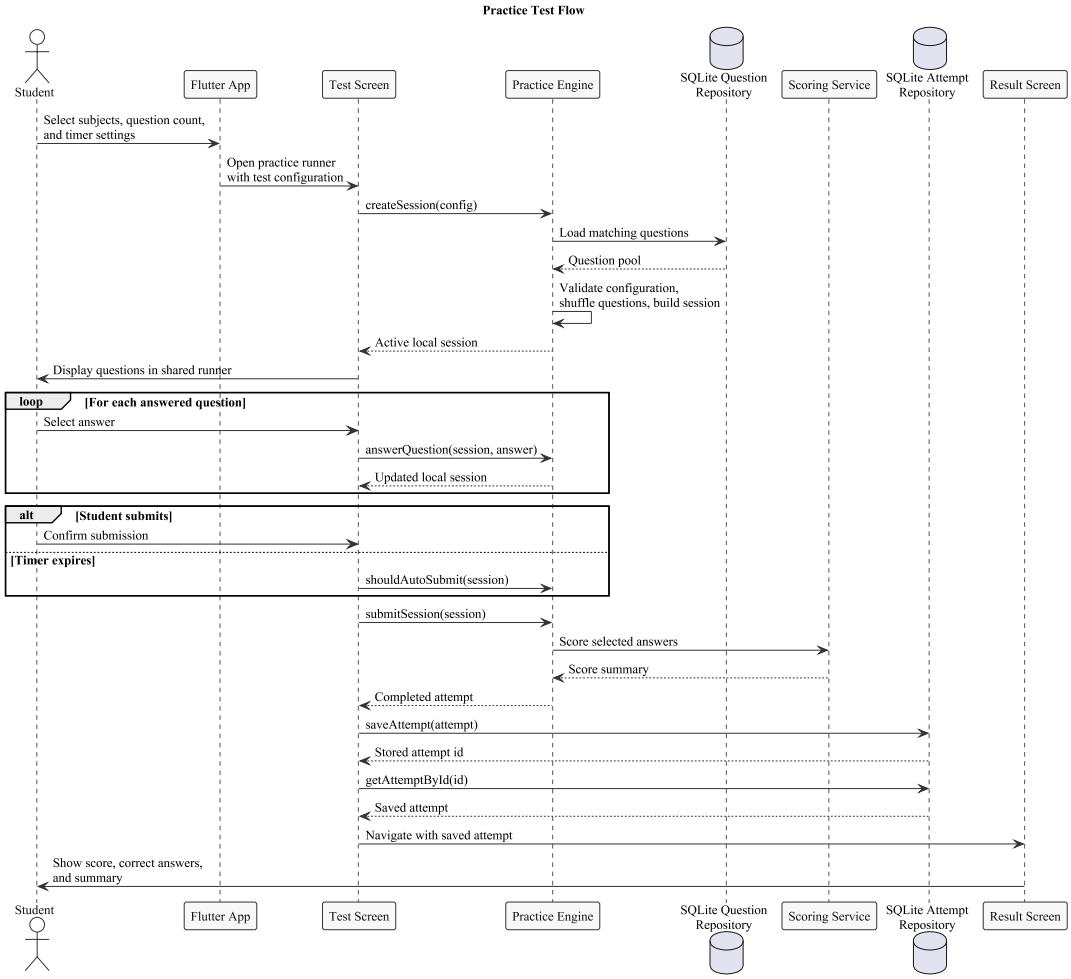


Figure 3.13: Sequence Diagram for Practice Test Flow

Figure 3.13 shows how a practice test is created and completed inside the mobile application. The student first selects the subjects, number of questions, and timer settings. The Flutter app opens the test screen with this configuration, and the test screen requests a local session from the practice engine. The practice engine then loads the matching question pool from the SQLite question repository, validates the configuration, shuffles the selected questions, and returns an active local session.

During the test, the shared runner displays each question and stores the selected answers in the active session. When the student submits the test, or when the timer expires, the test screen asks the practice engine to submit the session. The scoring service grades the selected answers locally, and the completed attempt is saved through the SQLite attempt repository. The result screen then presents the score, correct-answer summary, and attempt details to the student. This flow shows that ordinary practice tests do not depend on Firebase or a live connection, which supports the offline-first design goal of Exit 360.

Figure 3.14 presents the second sequence diagram, which describes the challenge-test flow. This workflow is more complex than ordinary practice because two students participate asynchronously, while the backend controls the trusted challenge state, question selection, sessions, grading, and final outcome.

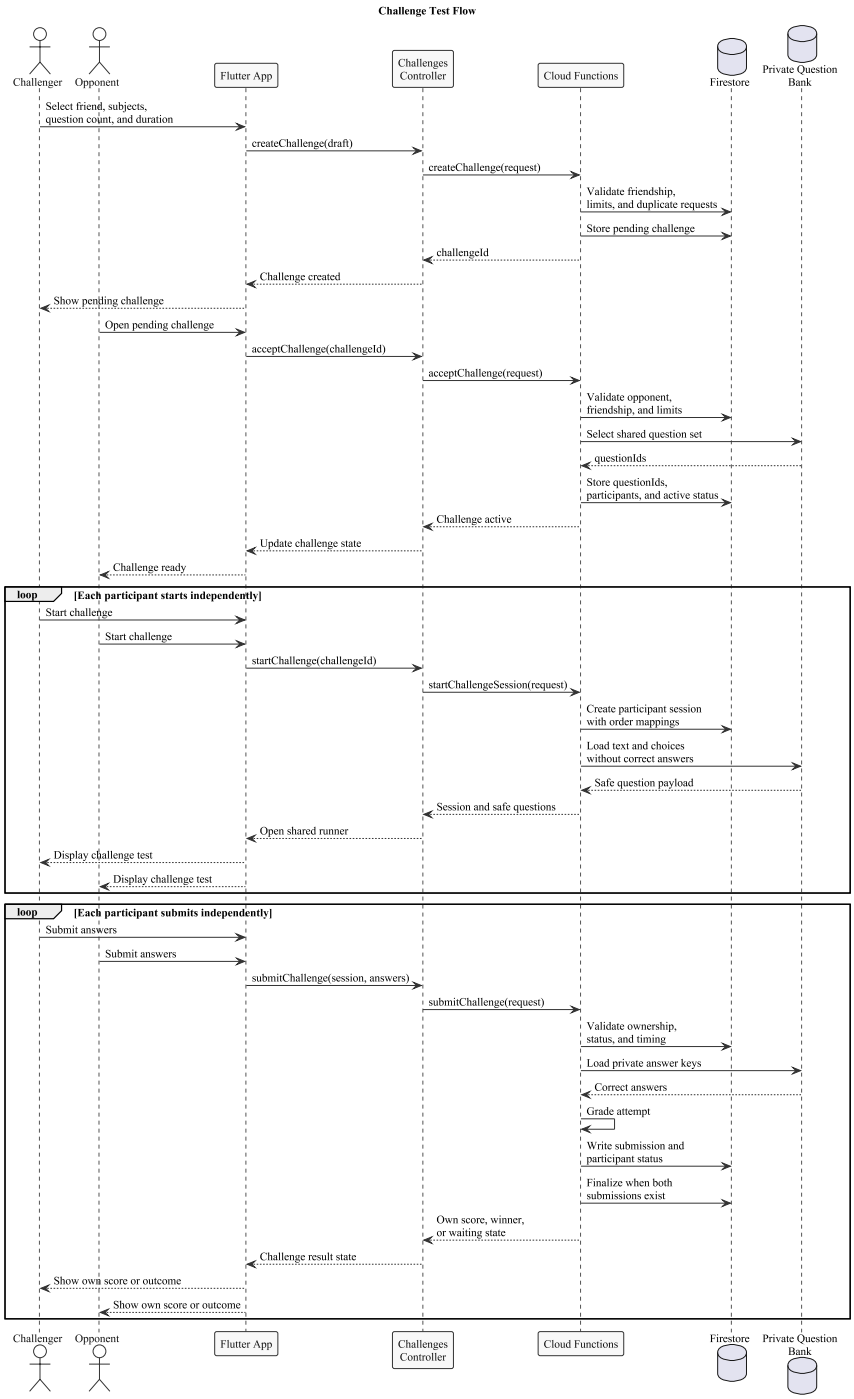


Figure 3.14: Sequence Diagram for Challenge Test Flow

Figure 3.14 begins when a challenger selects a friend, subject set, question count, and duration. The mobile app passes this draft to the challenges controller, which calls the backend function for challenge creation. The backend validates the friendship, weekly limits, and duplicate active requests before storing the challenge as a pending Firestore document. When the opponent accepts, the backend rechecks the trusted conditions, selects the shared question set from the private question bank, stores the locked question identifiers, and moves the challenge to the active state.

After activation, each participant may start and complete the test independently. For each student, the backend creates a separate test session with its own question and choice-order mappings, then returns only safe question text and choices to the mobile app. Correct answers are not sent to the client. When a participant submits, the backend validates ownership, session state, and timing, loads the private answer keys, grades the attempt, writes the official submission, and updates the participant state. If both participants have submitted, the backend finalizes the challenge by computing the winner or tie using score first and duration as the tie-breaker. This sequence shows why challenge tests require server-owned control even though the test-taking interface is reused from the shared runner.

Figure 3.15 presents the third sequence diagram, which describes the weekly-test flow. This workflow includes both the admin side and the student side because weekly tests are not ordinary local tests. They are planned through the internal admin panel, generated from the private question bank, attempted by students through the mobile app, and finally published through an official leaderboard snapshot.

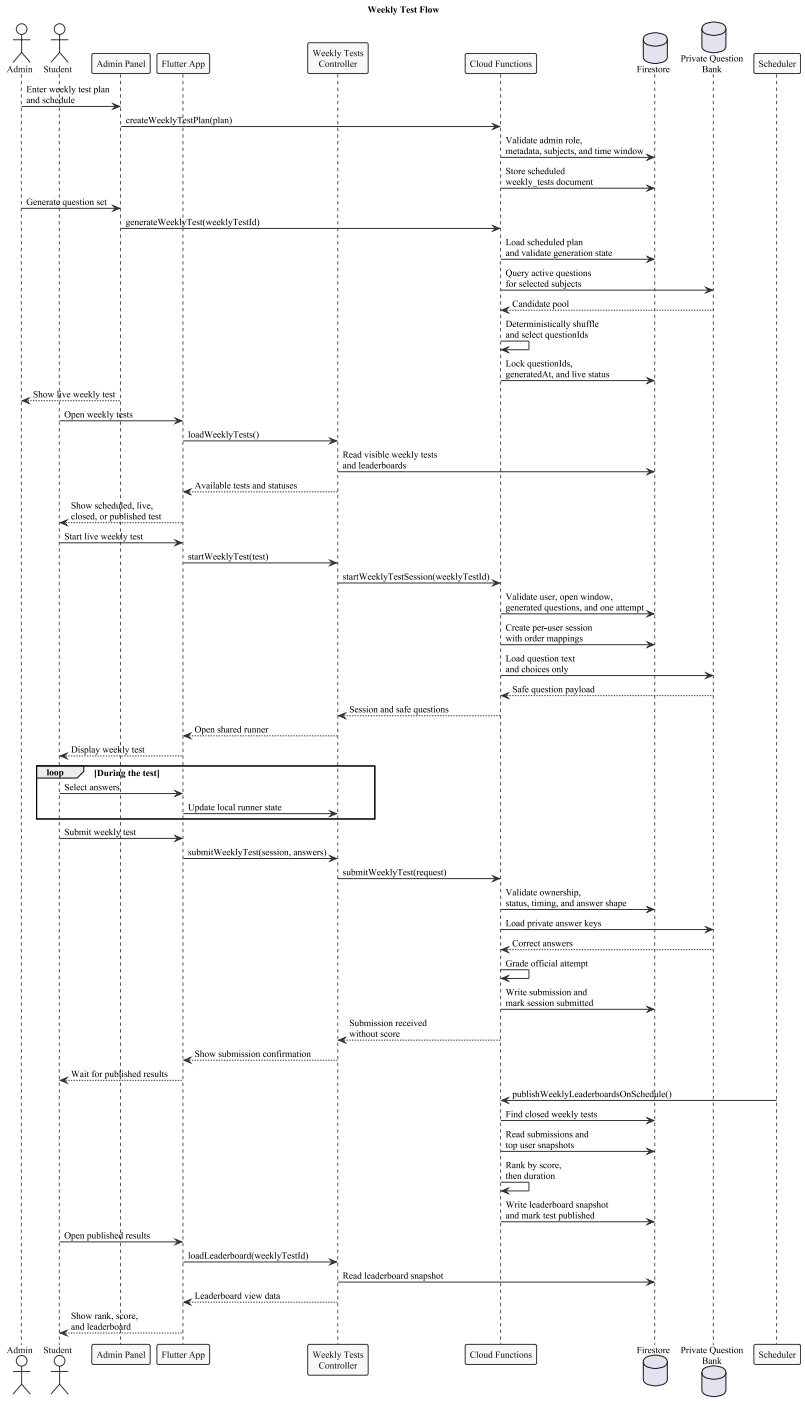


Figure 3.15: Sequence Diagram for Weekly Test Flow

Figure 3.15 starts with the administrator creating a weekly-test plan from the admin panel. The plan contains the test identity, subjects, question count, duration, and opening and closing times. The request is handled by Cloud Functions, which verifies that the user has admin privileges and validates the weekly-test metadata before storing the scheduled document in Firestore. When the question set is generated, the backend loads the scheduled plan, queries the active private question bank for the selected subjects, deterministically selects the required question identifiers, and locks them into the weekly-test document. This step is important because all students must compete on the same trusted question set, even if the order of questions and choices is different for each user.

On the student side, the mobile app loads visible weekly tests and published leaderboards from Firestore. When a student starts a live weekly test, the weekly tests controller calls the backend function responsible for creating the session. The backend validates the user, the test window, the generated question set, and the one-attempt rule. It then creates a per-user test session with question and choice-order mappings and returns only safe question text and choices to the app. As with challenge tests, correct answers are never sent to the client during the test.

When the student submits, the backend validates ownership, timing, session state, and answer shape before loading the private answer keys and grading the attempt. The official submission is written to Firestore, and the session is marked as submitted. Unlike local practice tests, the weekly-test submission does not immediately return the score to the student. The result is held until the weekly test closes and the scheduled publishing process ranks all submissions by score and then by duration. The final leaderboard is written as an immutable snapshot, and the weekly test is marked as published. This design keeps competitive results consistent, prevents answer exposure, and avoids the cost and complexity of a continuously updating leaderboard.

3.3.3 Architectural Design

The architecture of Exit 360 follows an offline-first mobile design with a cloud-backed online platform. The mobile application is responsible for the student learning experience, local practice tests, attempt history, and offline statistics. The backend is responsible for trusted online workflows such as authentication, competitive sessions, server-side grading, leaderboard publishing, admin operations, and notification support.

Figure 3.16 shows the main layers of the system. The Flutter mobile application contains the presentation layer, application logic, online feature clients, and

the shared test runner used by practice, weekly, and challenge tests. For offline preparation, the application uses a bundled SQLite question bank and local storage for attempts, answers, history, and statistics. This local layer allows students to continue practicing even when Firebase services are not reachable.

The online platform is built around Firebase Authentication, Cloud Firestore, Cloud Functions, and Firebase Cloud Messaging support. Authentication provides user identity for students and admins. Firestore stores profiles, friends, weekly tests, challenges, sessions, submissions, and leaderboard snapshots. Cloud Functions form the trusted backend layer, meaning that competitive question selection, session creation, answer grading, challenge state updates, and leaderboard publication are controlled by the server rather than by the mobile client. Firebase Cloud Messaging is included in the architecture as notification support for weekly tests, challenges, and similar reminders, although it is not treated as a core dependency for offline practice.

The internal admin panel is separated from the student-facing mobile app. It is used by authorized admins to manage weekly tests, inspect online activity, review question summaries, and trigger protected backend operations through Cloud Functions. This separation keeps the architecture clear. The mobile app supports learning and interaction, while the backend owns trusted data, competitive integrity, administration, and online operational workflows.

The complete architectural overview is presented in Figure 3.16.

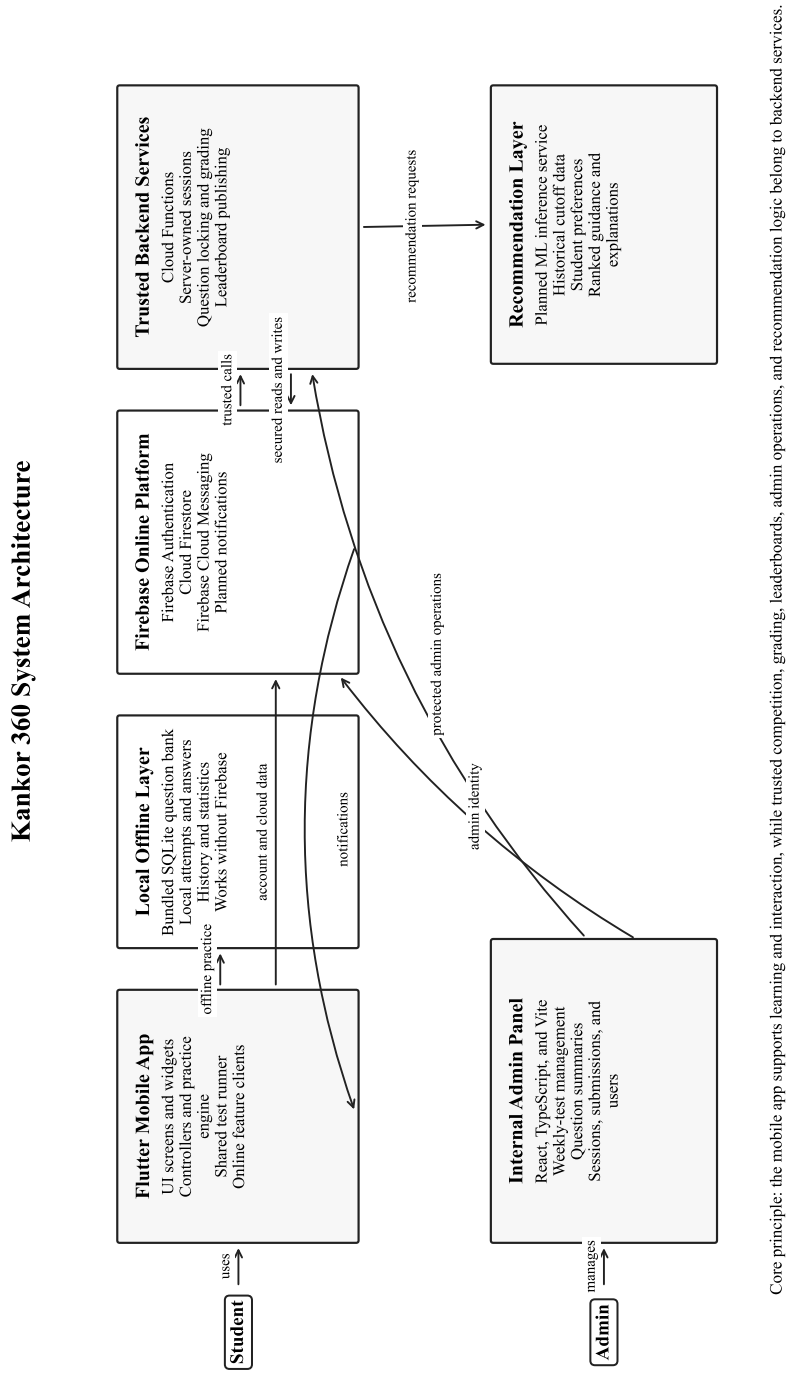


Figure 3.16: Architecture Diagram

As shown in Figure 3.16, the architecture separated local preparation features from trusted online workflows and admin operations.

3.3.4 Database Design

The offline part of Exit 360 uses a local SQLite database to support practice tests without requiring a continuous internet connection. This database stores the academic structure used by the question bank, the questions and answer choices shown during practice, and the locally cached summary statistics used for progress tracking. The design in this subsection focuses only on the local SQLite database. The Firestore data model is not included here because it belongs to the online and cloud-backed part of the system.

Figure 3.17 shows the entity-relationship diagram for the local database. The diagram uses crow's foot notation to show the cardinality between entities. Solid relationships represent direct foreign-key relationships in the SQLite design, while dashed relationships represent logical target relationships that are stored through target fields rather than through a single fixed foreign key.

As shown in Figure 3.17, the central learning content begins with `SchoolClass` and `SubjectGroup`. A subject belongs to one school class and one subject group, which allows questions to be organized by both exam content level and Exit Exam subject category. Each `Subject` can contain many `Question` records, and each question has one or more `AnswerChoice` records. This structure keeps the question bank separated from test history, which makes it easier to reuse the same questions in different practice sessions.

The diagram also includes the entities used to keep local practice history and summaries. `TestAttempt` represents completed practice attempts. Since an attempt can target either a subject or a subject group, this connection is shown as a logical relationship in the ERD rather than a direct foreign-key relationship. `SubjectStats` and `GroupStats` represent cached summaries for subject-level and group-level progress so that the app can show recent and best performance quickly without recalculating all past attempts each time.

The entity relationships described above are summarized in Figure 3.17.

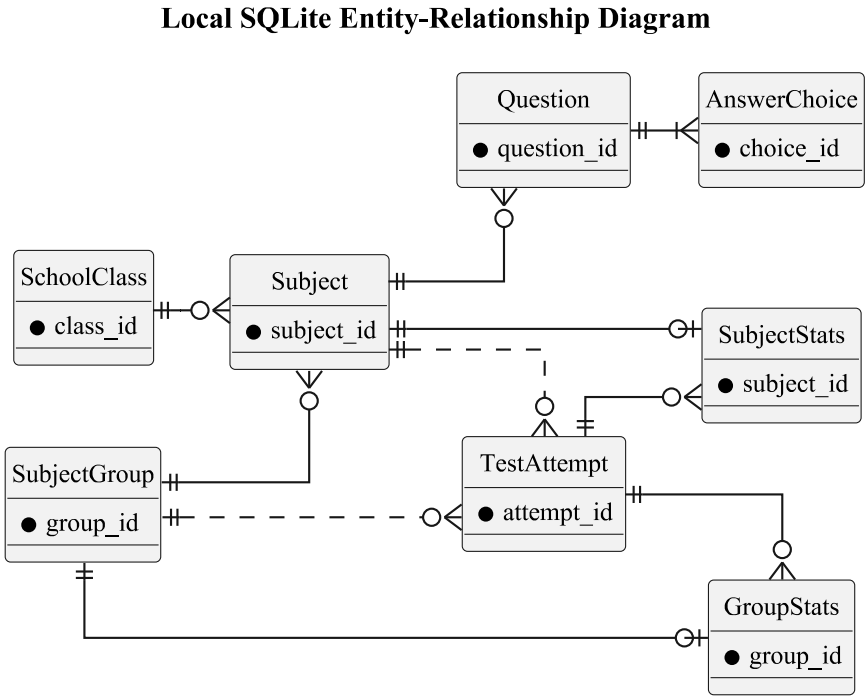


Figure 3.17: Local SQLite Entity-Relationship Diagram

Figure 3.18 presents the relational schema for the same local SQLite database. Unlike the ERD, this figure focuses on how the entities are stored as tables. It lists the main columns, data types, primary keys, foreign keys, uniqueness rules, default values, and important check constraints. This gives a more implementation-oriented view of the local database while the relationship structure remains visible in Figure 3.17.

As shown in Figure 3.18, the schema separates reference data, question-bank data, attempt history, and cached statistics. `SchoolClass`, `SubjectGroup`, and `Subject` define the academic structure. `Question` and `AnswerChoice` store the reusable question content and possible answers. `TestAttempt` stores the result of each completed local practice test, while `SubjectStats` and `GroupStats` store precomputed summaries for quick progress display. The use of simple SQLite data types keeps the local database lightweight and suitable for bundling with the mobile application.

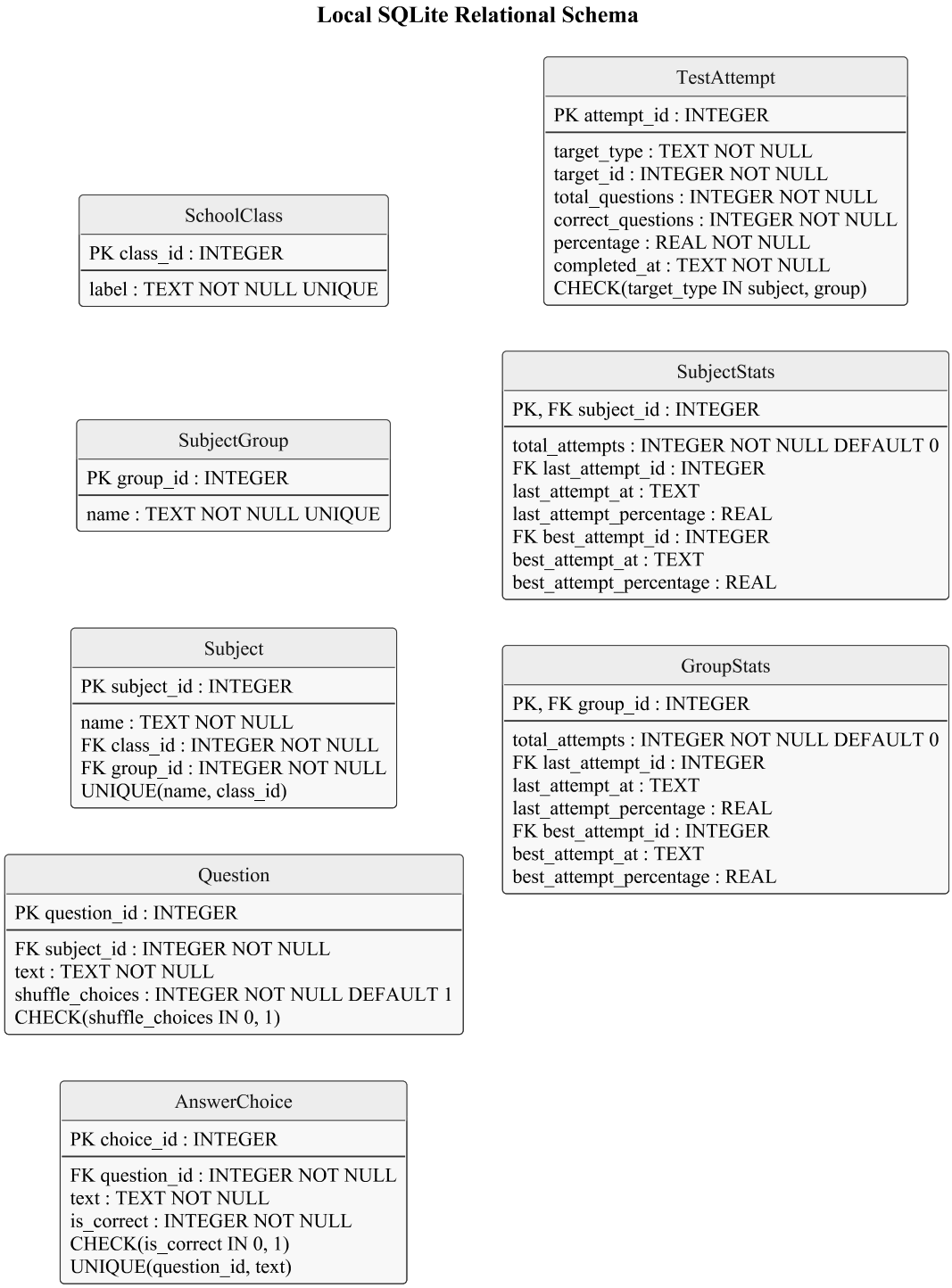


Figure 3.18: Local SQLite Relational Schema

Figure 3.18 shows how the local SQLite design was translated into concrete tables, columns, keys, and constraints. This schema view complemented the entity-relationship diagram by showing the implementation-level structure used by the mobile application's local database.

3.4 Implementation

This section summarizes how the proposed Exit 360 design was translated into working software. It describes the development environment, the main implementation phases, and the key modules created for the mobile application, Firebase backend, and internal admin panel. The purpose is to connect the earlier design models with the actual technical work completed during development.

3.4.1 Development Environment

The system was implemented inside a single monorepo so that the mobile application, Firebase backend, admin panel, thesis files, documentation, and supporting scripts could be developed and versioned together. Git and GitHub were used throughout the development process to keep a continuous history of changes and to make it easier to manage work across the different parts of the project.

The mobile application was developed using Flutter and Dart. Android Studio was used for Android SDK management, emulator support, and device-oriented testing, while Visual Studio Code was used as the main editor for Flutter, TypeScript, LaTeX, and project documentation. The local offline database was implemented with SQLite and accessed from the Flutter app through the sqflite package. This supported offline question retrieval, test attempts, result storage, history, and statistics without depending on a live internet connection.

The online layer was implemented with Firebase services. Firebase Authentication was used for user identity, Cloud Firestore was used for online application data, and Firebase Cloud Functions were used for trusted backend workflows such as online test sessions, grading, challenge operations, leaderboard publishing, and admin-side actions. The Firebase Emulator Suite was used during development to validate backend behavior, callable functions, and security-rule behavior locally before relying on deployed services.

The internal admin panel was developed as a separate React, TypeScript, and Vite web application. It was kept separate from the student-facing Flutter app because it supports privileged operational workflows such as preparing weekly tests, inspecting online activity, and triggering protected backend actions.

Python was used for data processing and supporting scripts, while LaTeX was used to prepare the final thesis document.

The principal technologies and versions used across these implementation layers are summarized in Table 3.1. The table provides a compact implementation record, while the paragraphs following it explain the purpose of each technology within Exit 360.

Table 3.1: Technology Stack

Layer	Technology	Version
Frontend	Flutter	3.27.3
Frontend language	Dart	3.6.1
Local storage	SQLite	3.46.1
Data access	sqflite	2.3.3
Authentication	Firebase Authentication	6.1.4
Cloud database	Cloud Firestore	6.1.2
Backend	Firebase Cloud Functions	7.0.0
Backend language	TypeScript	5.7.3
Admin frontend	React	19.2.6
Admin tooling	Vite	8.0.12
Admin language	TypeScript	6.0.2
Alerts	Firebase Cloud Messaging	Not applicable
Testing	Firebase Emulator Suite	Not applicable

As listed in Table 3.1, Flutter formed the frontend framework of the mobile application. It enabled the student-facing preparation features to be developed from one codebase and provided a practical path for possible expansion beyond Android in the future. Its efficient interface rendering, mature package ecosystem, and established support for mobile development made it suitable for building the complete application within the capstone timeline.

Dart was used as the frontend programming language. It supported the implementation of application screens, navigation, state handling, test logic, local database operations, and communication with the Firebase-backed online services. Its direct integration with Flutter and support for typed, asynchronous

programming reduced the need for additional language bridges across these features.

SQLite provided the local storage layer for the offline preparation experience. It stored the prebundled question bank, practice attempts, selected answers, results, history, and statistics so that the main preparation workflow could remain available without an internet connection. Its lightweight, embedded, and relational structure was appropriate for organized on-device data that did not require a separate database server.

The sqflite package connected the Flutter application to SQLite. It provided the data-access operations required to retrieve questions, save attempts and answers, calculate local statistics, and manage other offline practice records on the device. It was suitable because it exposed SQLite operations through a Flutter-compatible interface and allowed the offline layer to remain within the same Dart codebase.

Firebase Authentication provided identity management for the online parts of the system. It supported account access for student profiles and competitive workflows, as well as authenticated access to the internal admin panel. Its integration with the other Firebase services avoided the need to build and maintain a separate authentication server while still supporting protected backend operations.

Cloud Firestore served as the cloud database for online application data. It stored profiles, friend relationships, weekly tests, challenge records, sessions, submissions, results, and leaderboard snapshots that needed to be shared or synchronized between users. Its document model, real-time synchronization, security rules, and direct integration with Firebase Authentication and Cloud Functions suited the shared and event-driven nature of these workflows.

Firebase Cloud Functions provided trusted server-side execution for sensitive online workflows. It handled operations such as session creation, protected question delivery, grading, challenge processing, result generation, leaderboard publication, and privileged administrative actions. It was necessary because these operations could not safely rely on client-side logic, and its managed execution model integrated directly with the project's Firebase data and identity services.

TypeScript version 5.7.3 was used for the Firebase backend. Its type checking helped maintain the request and response contracts of callable functions and reduced errors across the trusted backend workflows. This was especially useful for keeping the data exchanged between the Flutter app, admin panel, and

backend consistent as the online features expanded.

React provided the component-based frontend of the internal admin panel. It supported the dashboard, weekly-test administration, question-bank summaries, user and submission inspection, and other operational pages used by administrators. Its reusable component model and mature web ecosystem made it appropriate for a data-oriented interface containing forms, filters, tables, charts, and modal dialogs.

Vite supplied the development and build tooling for the React admin panel. It supported a fast local development workflow and produced the web application bundle used for the administrative interface. Its lightweight configuration and rapid development server reduced setup overhead for the separate web application.

TypeScript version 6.0.2 was used in the admin panel. It added type checking to interface state, backend request models, and Firebase function calls, which made the administrative workflows easier to maintain reliably. Using typed models was particularly important because the panel handled privileged operations and shared backend contracts with Cloud Functions.

Firebase Cloud Messaging was selected for alerts related to weekly tests, challenge activity, leaderboard publication, reminders, and similar online events. It allowed these time-sensitive updates to be delivered without making the offline preparation workflow dependent on notifications. Its compatibility with the existing Firebase backend also allowed notification events to be triggered from trusted server-side workflows without introducing a separate messaging service.

The Firebase Emulator Suite supported local testing of Cloud Functions, Firestore behavior, authentication-related flows, and security rules. It allowed trusted backend workflows to be validated in a controlled environment before they were tested against deployed Firebase services. This reduced the risk of affecting production data and made repeated backend and security testing faster during development.

3.4.2 Implementation Phases

The implementation followed the same broad iteration structure described in the development methodology. Each iteration focused on turning one part of the design into working software while keeping the system organized around the offline-first architecture.

The first implementation phase focused on the offline mobile foundation. This included the Flutter project structure, navigation flow, reusable interface components, bundled SQLite question bank, subject-based practice tests, mixed practice tests, automatic grading, result display, attempt history, and local statistics. This phase was important because it produced the core preparation experience that students can use even when internet access is unavailable.

The second phase focused on the online student features and Firebase-backed workflows. User authentication, profile-related data, Firestore integration, online sessions, weekly tests, challenge workflows, and leaderboard-related backend logic were implemented around the principle that competitive integrity should belong to the backend. The mobile app collects user actions and displays results, while Cloud Functions control trusted operations such as session creation, answer validation, scoring, and publication of competitive results.

The third phase focused on the internal admin panel. The admin panel was implemented as a developer and operator-facing web application rather than as part of the mobile app. Its purpose was to support controlled management of online workflows, especially weekly tests, question summaries, operational inspection, and protected backend actions. This helped keep administrative functions away from the student-facing interface while still making them usable during development and testing.

3.4.3 Key Modules Implemented

The Flutter mobile app was implemented as the main student-facing product. Its key modules include practice navigation, subject selection, test configuration, the test-taking screen, automatic grading, result review, attempt history, and progress statistics. The offline modules use the local SQLite database so that ordinary practice workflows can continue without Firebase access.

The local data module stores the bundled academic structure, questions, answer choices, completed attempts, and cached statistics. This module supports the app's offline-first behavior by keeping the most important preparation features available on the device. It also separates reusable question content from attempt history, which makes the practice flow easier to maintain.

The Firebase backend module supports online workflows that need trusted server-side control. Cloud Functions handle operations where the mobile client should not be authoritative, including competitive test session creation, grading, challenge state changes, leaderboard publishing, and protected admin operations. Firestore stores online data such as profiles, social records, weekly

tests, challenges, sessions, submissions, and leaderboard snapshots.

The admin panel module provides an internal interface for privileged workflows. It connects to Firebase services and calls protected backend functions instead of directly modifying trusted competitive state from the browser. This keeps sensitive operational work separate from the student-facing mobile application.

3.4.4 Challenges and Solutions

During development, several practical challenges appeared because the system had to support both offline preparation and online competitive workflows. The challenges affected connectivity, competitive integrity, content organization, administration, cloud deployment, cost control, and testing, and each required a solution suited to the project's architecture and constraints.

Unreliable internet access was a central challenge because the main preparation experience needed to remain useful under limited connectivity. The core practice workflow was therefore designed as an offline-first Flutter feature using a bundled SQLite question bank and local storage for test attempts.

Keeping online competitive tests fair and resistant to tampering required a different approach. Trusted operations such as session creation, question locking, grading, and leaderboard publishing were moved to Firebase Cloud Functions instead of being controlled by the mobile client.

Correct answers also had to be protected while weekly and challenge tests were in progress. Safe question payloads were therefore used for these sessions, allowing the application to receive the question text and answer choices without receiving the correct-answer indexes.

The different structures required for Exit Exam practice created another implementation challenge. Configurable practice-test generation was developed to support selected subjects, mixed-subject selection, clinical and paraclinical groupings, question-count limits, and fallback behavior when fewer questions than requested were available.

Providing useful progress feedback without repeatedly performing expensive calculations also required a focused solution. Local statistical summaries were added for recent attempts, best performance, average score, trend direction, strongest subject, and weakest subject.

Privileged online workflows could not safely be exposed through the student-facing application. A separate internal admin panel was therefore connected to protected backend functions for weekly-test planning, question-set generation,

leaderboard publishing, and broader platform inspection.

The limited local question bank presented a content-related challenge. The SQLite schema and application logic were structured so that the current Dari question bank could be expanded later, while the preparation of a comprehensive bilingual Dari and Pashto question bank was left for future expert-supported work.

Routine administration initially risked requiring direct manipulation of Firestore data. Common operational workflows were consequently added to the admin panel, reducing the need to edit Firestore documents manually when managing weekly tests and published results.

Backend reads, writes, and Firebase usage costs also needed to be controlled. Backend-generated leaderboard snapshots, asynchronous challenge workflows, and offline local practice were used so that the system would not depend on continuous real-time cloud updates.

Deploying Firebase Cloud Functions from Afghanistan introduced a separate operational difficulty. The Firebase project had to be upgraded from the free Spark plan to the pay-as-you-go Blaze plan because Cloud Functions deployment required billing. Since Google Cloud Billing did not support Afghan payment methods, a supported foreign payment card was used to complete the deployment setup.

Finally, the growing Flutter codebase increased the number of edge cases that needed to be checked. Automated Flutter tests were added for scoring, practice generation, statistics, models, controllers, widget states, authentication flows, weekly tests, challenges, and review screens. These automated tests were supported by manual checks using the Firebase Emulator Suite.

3.5 Testing

This section explains how the implemented Exit 360 system was tested and evaluated within the project scope. Unit testing verified core domain logic, data models, controllers, and selected interface states, while integration testing examined the connections among the Flutter application, local data layer, Firebase services, and admin panel. User acceptance evidence was gathered through the iteration-based prototype reviews conducted with target users and fellow developers. Together, these activities provided evidence that the principal offline, online, and administrative workflows behaved as expected.

3.5.1 Unit Testing

Automated tests were added for important Flutter domain logic, Firebase data models, controllers, and selected widget flows. The tests were written using Flutter's testing framework. They cover core behavior such as practice-test generation, scoring, statistics, authentication screens, Firebase model parsing, weekly-test helpers, challenge helpers, friends and competition screens, profile editing, test-taking behavior, and UI state handling. The following tables present representative automated test cases from this test suite in four functional groups.

Each test case uses the same five fields. The Test Case field identifies the behavior being verified, while Input describes the data, state, or user interaction supplied to the test. Expected Result states the behavior that should occur when the implementation works correctly, and Actual Result records the behavior observed when the automated test was executed. Status indicates whether the observed result matched the expected result.

The first group covers the core offline preparation logic. Table 3.2 presents tests for scoring, edge cases, practice-session generation, invalid configurations, and progress statistics.

Table 3.2: Core Practice and Scoring Unit Test Cases

UT-01	
Test Case	Scoring service calculates mixed correct, incorrect, and unanswered answers
Input	Three questions with one correct answer, one incorrect answer, and one unanswered question
Expected Result	The summary contains correct, incorrect, unanswered, answered, percentage, and per-subject totals accurately
Actual Result	The scoring service returned one correct, one incorrect, one unanswered, and the expected per-subject summary
Status	Pass
UT-02	
Test Case	Scoring service handles edge-case result sets

Input	All-correct, all-incorrect, all-unanswered, and zero-question test data
Expected Result	Each edge case returns a stable score summary without calculation errors
Actual Result	The service returned 100 percent for all-correct answers, 0 percent for incorrect or unanswered answers, and a stable empty summary for zero questions
Status	Pass

UT-03	
Test Case	Practice engine creates a valid practice session
Input	Five available subject questions with a request for three questions
Expected Result	The engine creates a session with the requested number of questions, starts at the first question, and records no selected answers initially
Actual Result	A successful session was created with three questions, current index zero, and zero answered questions
Status	Pass

UT-04	
Test Case	Practice engine rejects invalid or unavailable question configurations
Input	Invalid timed configuration, empty question repository, and a request larger than the available question count
Expected Result	Invalid configuration and no-question states return failure results, while short question pools use all available questions safely
Actual Result	The engine returned invalid configuration and no-questions failures where appropriate and used the shortfall fallback when fewer questions were available
Status	Pass

UT-05	
Test Case	Practice statistics summarize progress correctly
Input	Empty attempt history, one-attempt history, and multiple attempts with subject scores
Expected Result	The statistics service returns empty details when needed and calculates average score, best score, latest score, trend, strongest subject, and weakest subject correctly
Actual Result	The service returned the expected empty state and correctly summarized progress trends and subject strengths or weaknesses
Status	Pass

As shown in Table 3.2, the core practice tests passed across normal, invalid, limited-data, and empty-data conditions. Together, they confirmed that the application could generate valid practice sessions, handle unavailable configurations safely, calculate scores consistently, and derive the statistics required for progress feedback.

The second group evaluates identity handling and the data logic used by the online features. Table 3.3 covers authentication behavior, Firebase model parsing, weekly-test helpers, and challenge-state processing.

Table 3.3: Authentication and Online Workflow Unit Test Cases

UT-06	
Test Case	Authentication screens validate and route users correctly
Input	Sign-up form data, email verification state, profile completion fields, sign-in errors, and verified or unverified users
Expected Result	The screens show validation messages, create accounts, route unverified users to verification, route incomplete profiles to profile completion, and show mapped authentication errors

Actual Result	Widget tests confirmed the expected validation, routing, and localized error behavior
Status	Pass
UT-07	
Test Case	Firebase data models parse online documents correctly
Input	Sample user, reference collection, leaderboard, challenge, test session, and submission maps
Expected Result	Model objects should preserve identifiers, timestamps, participant states, question order, choice order, and scores
Actual Result	Model tests parsed the sample Firebase structures into the expected typed objects
Status	Pass
UT-08	
Test Case	Weekly-test helpers derive safe states and answer payloads
Input	Weekly-test status values, safe question payloads without correct answers, incomplete answers, and duration values
Expected Result	The helpers derive correct actions, avoid exposing correct answers, only build complete displayed-index answer payloads, and format durations correctly
Actual Result	The weekly-test tests confirmed correct status mapping, safe question mapping, answer-payload validation, duration formatting, and callable error mapping
Status	Pass
UT-09	
Test Case	Challenge helpers handle challenge states and role-aware actions

Input	Pending, active, completed, declined, canceled, and expired challenge states with challenger and opponent roles
Expected Result	The helpers should group challenges correctly, expose the proper actions for each role, and build complete answer payloads only when valid
Actual Result	The challenge tests confirmed correct status grouping, role-aware actions, displayed-index payload creation, and callable error mapping
Status	Pass

The results in Table 3.3 confirmed that authenticated users were routed according to verification and profile state, while Firebase documents and online workflow states were converted into the expected typed models and actions. The tests also confirmed that weekly and challenge answer payloads were validated without exposing correct answers.

The third group focuses on the student-facing flow from completing a practice attempt to reviewing its result and interacting with an active test. Table 3.4 presents the related model and widget test cases.

Table 3.4: Test-Taking and Result Interface Unit Test Cases

UT-10	
Test Case	Result and review screens present completed attempts correctly
Input	Completed practice attempt with summary counts, per-subject breakdown, and selected answers
Expected Result	The result screen displays total, correct, incorrect, unanswered, and per-subject values, and the review action opens the review screen with the same attempt
Actual Result	Widget tests confirmed the summary display, invalid-route fallback, and navigation from results to review
Status	Pass

UT-11	
Test Case	Active test session tracks navigation, answer count, and timed state
Input	Active test session with current index, selected answers, timed configuration, and expiry data
Expected Result	The session should report answered and unanswered counts, expose valid navigation state, and calculate remaining timed-session information correctly
Actual Result	Model tests confirmed stable answer counts, navigation getters, timed-session state, and remaining-duration behavior
Status	Pass

UT-12	
Test Case	Test-taking screen handles layout, answer selection, timer, submission, and exit confirmation
Input	Narrow portrait screen, long answer text, timed session data, answer taps, submit action, and exit action
Expected Result	Footer actions remain visible, long options do not overflow, selected answers are tracked, timed sessions show remaining time, and submit or exit confirmation dialogs appear when required
Actual Result	Widget tests confirmed responsive layout behavior, long option rendering, answer selection, timer display, submit confirmation, and discard confirmation
Status	Pass

As summarized in Table 3.4, the result, review, active-session, and test-taking interfaces behaved correctly across navigation, answer tracking, timing, responsive layout, submission, and exit-confirmation scenarios. These tests connected the underlying session state to the information and controls presented to the user.

The final group covers the wider social and competition interface, communication with callable backend functions, and common states across entry,

profile, history, and statistics screens. Table 3.5 presents these representative cases.

Table 3.5: Social, Competition, and General Interface Unit Test Cases

UT-13	
Test Case	Friends discovery and relationship actions behave correctly
Input	Username search text, current user identity, search results, incoming requests, outgoing requests, existing friends, and backend error codes
Expected Result	Search should normalize valid queries, avoid searching too early, hide the current user from results, show the correct relationship action, and map backend errors clearly
Actual Result	Controller and widget tests confirmed query normalization, debounced search behavior, relationship-state display, send, accept, decline, and cancel actions, and error mapping
Status	Pass
UT-14	
Test Case	Competition hub and weekly-test screens display online competition states
Input	Loading, error, empty, scheduled, live, closed, published, leaderboard, and challenge summary states
Expected Result	The competition screens should show appropriate loading, retry, empty, live-test, leaderboard, and challenge information based on controller state
Actual Result	Widget and controller tests confirmed competition hub summary behavior and weekly-test screen rendering for scheduled, live, closed, and published states
Status	Pass

UT-15	
Test Case	Cloud Function request and response models preserve backend contracts
Input	Weekly-test submission request, safe weekly questions, and challenge session payloads returned from callable functions
Expected Result	Request serialization should match the backend contract, and safe question responses should be parsed without exposing correct answers
Actual Result	Function service tests confirmed request serialization, safe weekly-question parsing, and challenge-session response parsing without correct-answer leakage
Status	Pass
UT-16	
Test Case	Entry, profile, practice history, and practice statistics screens handle common UI states
Input	Back navigation, unsigned competition access, profile edit fields, empty practice history, stored attempts, stored statistics, and load failures
Expected Result	Screens should route users based on authentication state, save allowed profile fields, show empty-state calls to action, render stored records, and show localized fallback messages on load failure
Actual Result	Widget tests confirmed entry routing, profile editing, history and statistics rendering, empty states, and localized failure fallbacks
Status	Pass

Table 3.5 shows that the broader application interfaces passed tests for friend discovery, relationship actions, competition states, backend request and response contracts, profile editing, stored practice information, empty states, and localized failure handling. Taken together, the four groups provided automated evidence across the main offline, online, and interface responsibilities of the Flutter application.

3.5.2 Integration Testing

Automated tests covered important contracts and boundary points, such as Cloud Function request serialization, safe question response parsing, Firebase model parsing, controller behavior, and widget state handling. These tests helped reduce the risk of breaking the connection between the Flutter app, data models, and backend-facing service layer.

Integration testing was then carried out through structured manual testing with the Firebase Emulator Suite and, later, with the deployed backend functions. During emulator-based testing, the mobile app and admin workflows were checked against local Firebase services so that authentication state, Firestore reads and writes, callable Cloud Functions, challenge operations, weekly-test flows, and security-rule behavior could be verified without depending on production data. This made it possible to test backend-controlled workflows in a safer local environment.

After the Firebase functions were deployed, the same major workflows were tested again against the real backend. This included account-related flows, online competition screens, weekly-test session behavior, challenge-related actions, callable function responses, and leaderboard or result-related data paths. The purpose of this stage was to confirm that the application still worked when connected to the deployed Firebase environment, not only when using local mocks or emulators.

The integration testing process showed that the most important system boundaries were the Flutter app with SQLite for offline practice, the Flutter app with Firebase Authentication for user identity, the Flutter app with Firestore and Cloud Functions for online workflows, and the admin panel with protected backend functions for operational tasks. Issues found during this process were handled through repeated manual checks, code fixes, and retesting of the affected flows.

3.5.3 User Acceptance Testing

User acceptance testing was handled through the iteration-based prototype reviews described in Section 3.2.4. Since each major iteration produced a working prototype, user acceptance was not treated as a single activity at the end of the project. Instead, feedback was collected after the offline preparation prototype, the online social and competition prototype, and the internal admin prototype had each reached a reviewable state.

This approach helped connect user acceptance with the actual development process. The first review focused on whether the offline practice workflow was understandable and useful for Exit Exam preparation. The second review focused on account-based online features such as weekly tests, leaderboards, friends, and challenges. The third review focused on whether the admin panel could support operational work without requiring direct Firestore manipulation. The detailed feedback, screenshots, and resulting refinements for these three prototype reviews are therefore presented earlier in Section 3.2.4.

The acceptance feedback confirmed that the main direction of the system was suitable, especially the offline-first preparation workflow and the addition of optional online competition features. It also identified practical improvements related to subject selection, review filters, attempt management, authentication polish, navigation consistency, Firestore rule issues, notifications, and admin-side question-bank management. These findings were used as user acceptance evidence and as guidance for later refinement of the system.

CHAPTER 4: RESULTS

This chapter presents the descriptive results of the Exit 360 questionnaire. The analysis was based on 47 valid responses collected from curative medicine and stomatology graduates, as well as final-year students who were preparing for the Afghan Medical Exit Exam. The chapter focuses on respondent profile, current preparation and access, feature interest and priorities, and the perceived impact of the proposed application.

The purpose of this analysis was to support the first research objective and the first research question stated in Chapter 1. Specifically, the results helped identify the existing challenges faced by Exit Exam candidates and clarified which system requirements and feature priorities needed to be considered when developing an Exit Exam preparation mobile application.

4.1 Questionnaire Results

The questionnaire results are presented in five subsections. The first subsection describes the respondents, while the next three subsections analyze preparation access, feature priorities, and expected impact. The final subsection summarizes the main findings and connects them to the requirements direction of Exit 360.

4.1.1 Respondent Profile

The respondent profile showed that the survey reached the intended population for this project. Figure 4.1 presents the study level of the respondents, showing the balance between graduates and final-year students from curative medicine and stomatology.

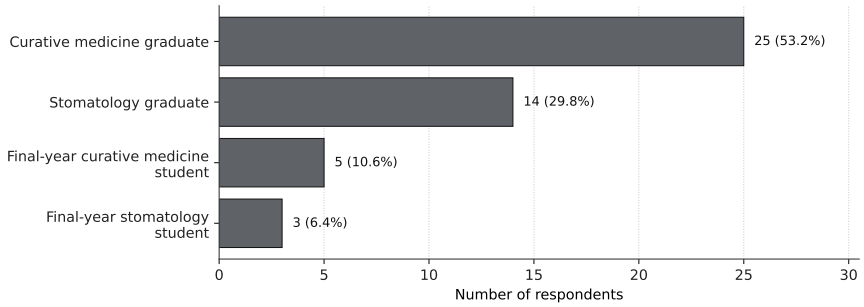


Figure 4.1: Respondent study level

As shown in Figure 4.1, curative medicine graduates formed the largest group, with 25 respondents or 53.2% of the sample. Stomatology graduates made up 14 respondents or 29.8%. The remaining respondents were final-year

students, including five final-year curative medicine students and three final-year stomatology students. This means that all respondents had a direct academic connection to the Exit Exam target population, rather than being a general student sample.

The respondents were also asked whether they had already taken the Afghan Medical Exit Exam. Figure 4.2 shows the distribution of respondents by their Exit Exam status.

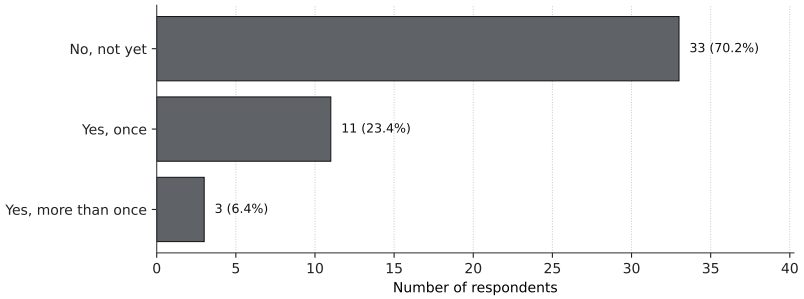


Figure 4.2: Exit Exam status of respondents

Figure 4.2 shows that 33 respondents, or 70.2%, had not yet taken the Exit Exam. Eleven respondents, or 23.4%, had taken it once, while three respondents, or 6.4%, had taken it more than once. This pattern was useful for requirements elicitation because most respondents were still preparing for the exam, while a smaller group had already experienced the exam process and could respond from direct examination experience.

The geographic background of the respondents was centered on Kabul, which was the province of origin for 26 respondents or 55.3% of the sample. The next largest group was from Ghazni, with six respondents or 12.8%, followed by Bamyan with three respondents or 6.4%. Smaller numbers of respondents came from Khost, Parwan, Daykundi, Kapisa, Baghlan, Ghor, and Balkh. This matched the practical collection context, but it also showed that the sample included respondents originally from several other provinces. All 47 respondents were male, so the findings should not be treated as representative of all Exit Exam candidates. The results were therefore useful for exploratory requirements analysis, but not for national generalization.

The sample also showed a strong connection to formal preparation activity. In total, 34 respondents, or 72.3%, reported that they were attending an Exit Exam preparation course or coaching program, while 13 respondents, or 27.7%, were not. This was important for the questionnaire because most participants were not responding from a distant or general interest in the exam. Instead, many of them

were actively involved in preparation when the questionnaire was answered, which made their responses useful for identifying practical preparation needs and feature priorities.

4.1.2 Current Preparation and Access

The questionnaire asked respondents how they were currently preparing for the Exit Exam. Since this was a multiple-response question, Figure 4.3 counts each selected method separately rather than treating the answers as exclusive categories.

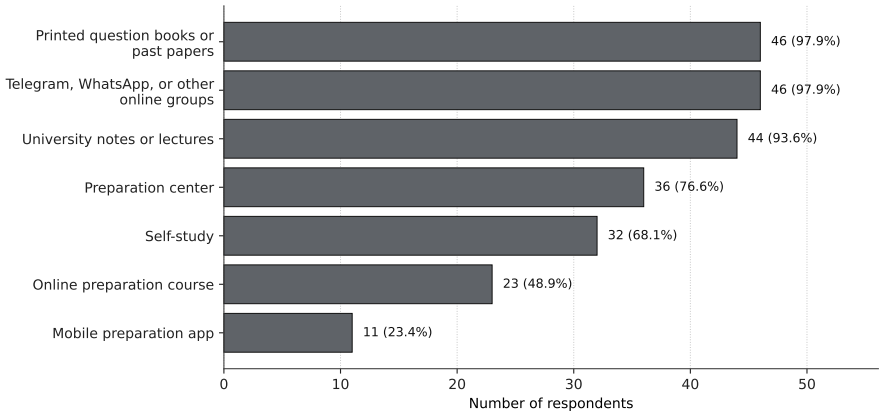


Figure 4.3: Current Exit Exam preparation methods

Figure 4.3 shows that preparation was not limited to one method. Printed question books or past papers and Telegram, WhatsApp, or other online groups were each selected by 46 respondents, or 97.9%. University notes or lectures were selected by 44 respondents, or 93.6%, while preparation centers were selected by 36 respondents, or 76.6%. Self-study was also common, with 32 respondents or 68.1%. Online preparation courses were used by 23 respondents, or 48.9%, while mobile preparation apps were selected by only 11 respondents, or 23.4%. This suggests that candidates relied on many fragmented resources, while mobile applications had not become the dominant preparation method in this sample.

The frequency of mock or practice test participation is shown in Figure 4.4. This figure helps show how regularly respondents practiced under test-like conditions.

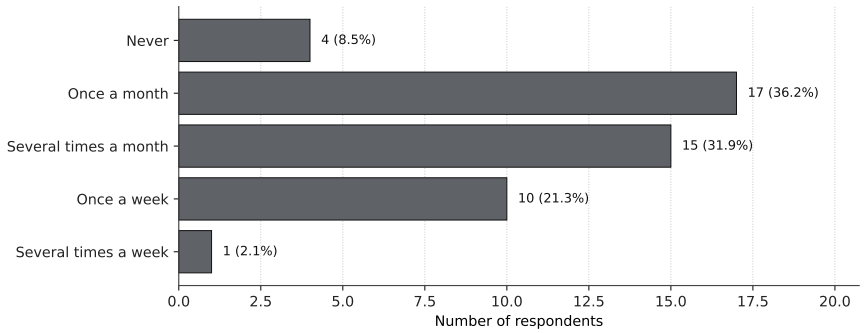


Figure 4.4: Frequency of mock or practice tests

As Figure 4.4 shows, 17 respondents, or 36.2%, reported taking mock or practice tests once a month, while 15 respondents, or 31.9%, took them several times a month. Ten respondents, or 21.3%, took practice tests once a week, and only one respondent took them several times a week. Four respondents, or 8.5%, reported that they never took mock or practice tests. These results suggest that practice testing existed among the respondents, but it was not frequent for everyone.

The questionnaire also asked how easy it was for respondents to access Exit Exam mock or practice tests on demand. Figure 4.5 presents those responses.

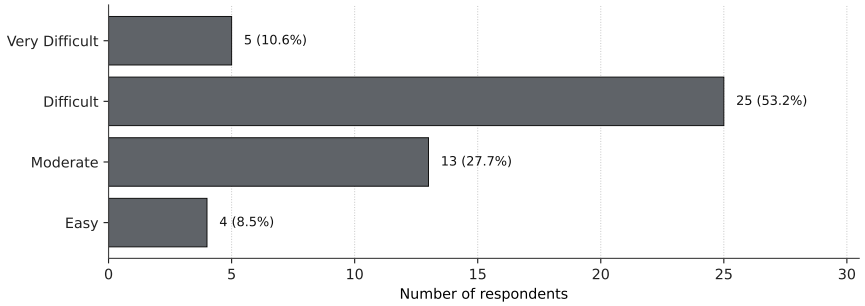


Figure 4.5: Ease of accessing practice tests on demand

Figure 4.5 shows a clear access problem. Twenty-five respondents, or 53.2%, described on-demand access as difficult, and five respondents, or 10.6%, described it as very difficult. Thirteen respondents, or 27.7%, rated access as moderate, while only four respondents, or 8.5%, described it as easy. This finding supported the need for a preparation tool that allowed candidates to start practice tests when they needed them, instead of depending only on scheduled or externally provided opportunities.

Internet availability was another important part of the access picture. Figure 4.6 shows how often respondents had internet access for studying.

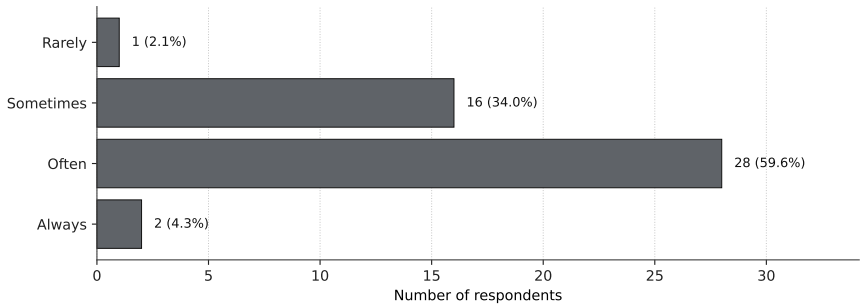


Figure 4.6: Internet access for studying

Figure 4.6 shows that 28 respondents, or 59.6%, often had internet access for studying, and two respondents, or 4.3%, always had access. However, 16 respondents, or 34.0%, only sometimes had internet access, and one respondent, or 2.1%, rarely had access. This result showed that internet access was available for many respondents, but not consistently enough to make a fully online system reliable for everyone.

Internet reliability was also measured separately. Figure 4.7 shows how respondents described the reliability of their connection.

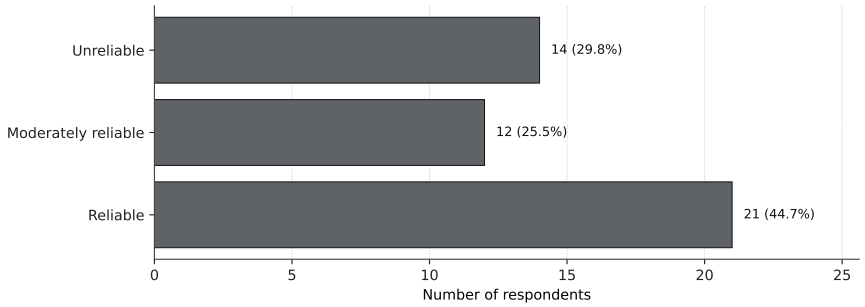


Figure 4.7: Reliability of internet connection

As shown in Figure 4.7, 21 respondents, or 44.7%, described their connection as reliable. However, 12 respondents, or 25.5%, described it as moderately reliable, and 14 respondents, or 29.8%, described it as unreliable. This result supported the design decision to keep core preparation features available offline, while using internet-based features as an additional layer when connectivity was available.

The importance of offline access was directly measured in the questionnaire. Figure 4.8 presents how respondents rated offline access to practice tests.

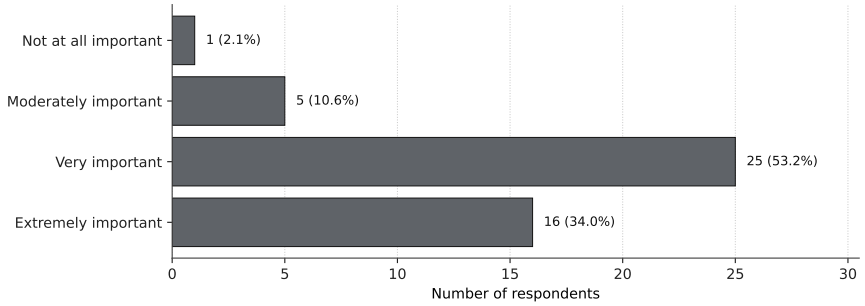


Figure 4.8: Importance of offline practice-test access

Figure 4.8 shows that offline access was strongly valued. Sixteen respondents, or 34.0%, rated offline access as extremely important, and 25 respondents, or 53.2%, rated it as very important. When the five respondents who rated it moderately important are included, 46 respondents, or 97.9%, considered offline access at least moderately important. This provided strong support for the offline-first design of Exit 360.

The questionnaire also asked whether respondents had ever used an Exit Exam or medical exam preparation app. Figure 4.9 shows the result.

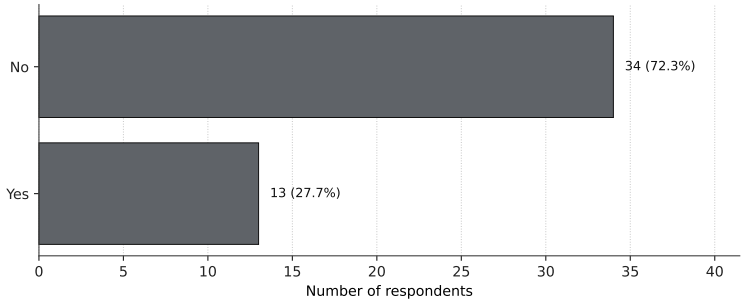


Figure 4.9: Prior use of Exit Exam or medical exam preparation apps

As Figure 4.9 shows, 34 respondents, or 72.3%, had not used an Exit Exam or medical exam preparation app, while 13 respondents, or 27.7%, had used one. Among those who named an app, eight mentioned EXIT Medical Exam Prep, and five mentioned MedQuiz, also known as Azmoonak Tebi. This result showed that local medical exam preparation apps had reached some respondents, but app-based preparation was still not universal in the sample.

4.1.3 Feature Interest and Priorities

The questionnaire asked respondents to rate the usefulness of six proposed Exit 360 features. Figure 4.10 presents the mean usefulness score for each feature on a five-point scale.

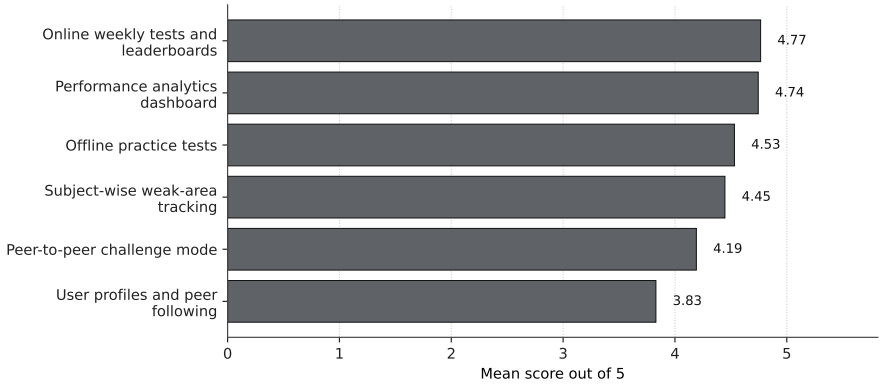


Figure 4.10: Mean usefulness rating of proposed Exit 360 features

Figure 4.10 shows that all proposed features received positive ratings. Online weekly competitive tests and leaderboards received the highest mean score, 4.77 out of 5. Performance analytics followed closely with a mean of 4.74. Offline practice tests had a mean score of 4.53, and subject-wise weak-area tracking had a mean score of 4.45. Peer-to-peer challenge mode received a mean score of 4.19. User profiles and peer following had the lowest mean score among the six features, 3.83, but it still remained positive. In terms of very useful or extremely useful responses, online weekly tests and leaderboards, performance analytics, and offline practice tests were each selected by 44 respondents, or 93.6%.

The priority-ranking question gave a more forced view of respondent preferences. Figure 4.11 shows the weighted top-three ranking, where a first-rank feature received three points, a second-rank feature received two points, and a third-rank feature received one point.

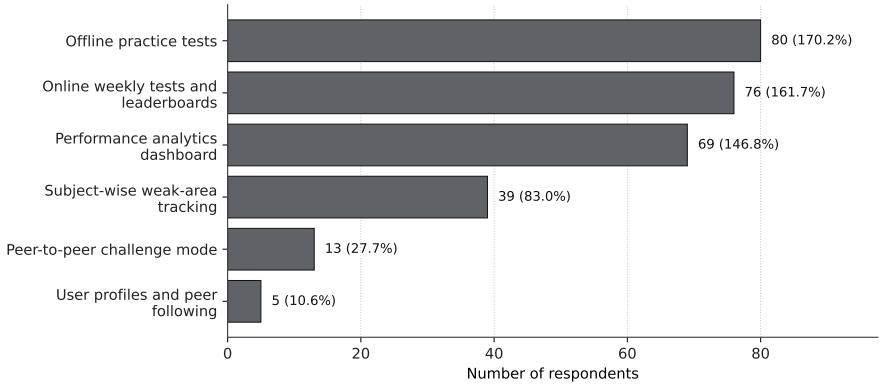


Figure 4.11: Top-three feature priority ranking

Figure 4.11 shows that offline practice tests ranked first with 80 weighted

points, 35 total mentions, and 18 first-rank votes. Online weekly tests and leaderboards ranked second with 76 points, 37 mentions, and 13 first-rank votes. Performance analytics ranked third with 69 points and 37 mentions. Subject-wise weak-area tracking received 39 points, peer-to-peer challenge mode received 13 points, and user profiles and peer following received five points. Taken together, Figures 4.10 and 4.11 showed that respondents valued both the offline preparation core and the online competition layer, while analytics and weak-area tracking were also important requirements.

The questionnaire also asked respondents how likely they would be to use an Exit Exam preparation mobile app with the proposed features. Figure 4.12 presents this adoption-likelihood result.

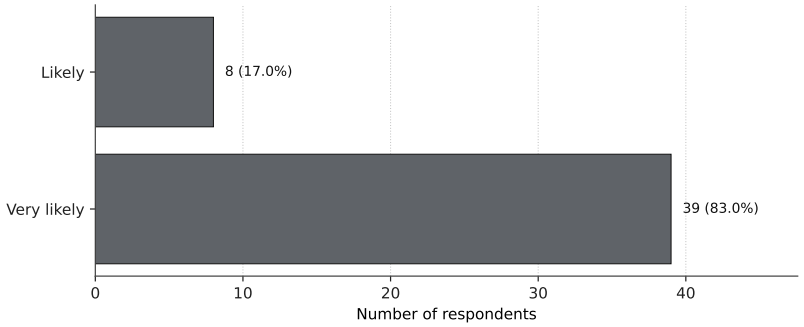


Figure 4.12: Likelihood of using the proposed Exit Exam preparation app

Figure 4.12 shows a strongly positive adoption response. Thirty-nine respondents, or 83.0%, said they were very likely to use such an application, while eight respondents, or 17.0%, said they were likely to use it. No respondent selected a neutral or negative option. This supported the practical relevance of developing an Exit Exam preparation mobile application with the proposed feature set.

4.1.4 Perceived Impact

The perceived-impact section measured how respondents expected the proposed application to support their preparation. Figure 4.13 presents the mean agreement scores for the seven impact statements.

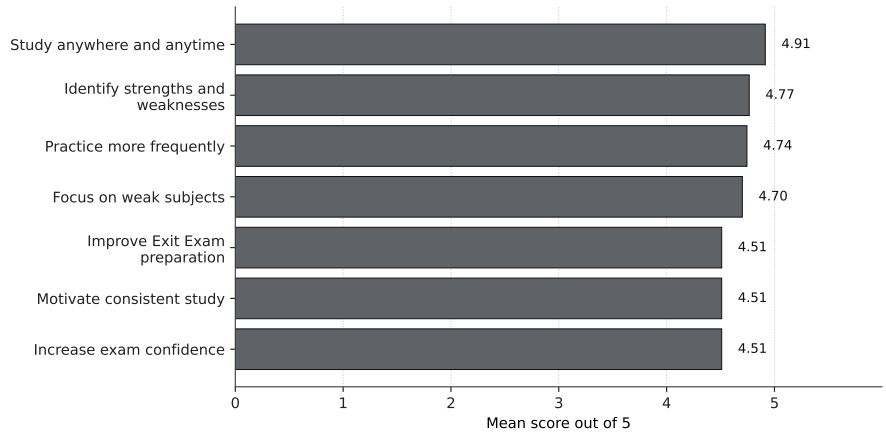


Figure 4.13: Mean agreement with expected app impact

Figure 4.13 shows that respondents expected the application to support several important parts of their preparation. The highest mean score was for studying anywhere and anytime, with a mean of 4.91. Identifying strengths and weaknesses had a mean score of 4.77, and practicing more frequently had a mean score of 4.74. Focusing on weak subjects had a mean score of 4.70. Improving Exit Exam preparation, motivating consistent study, and increasing exam confidence each had a mean score of 4.51.

The results in Figure 4.13 also showed very high agreement overall. All 47 respondents agreed or strongly agreed that the app would improve their preparation, help them practice more frequently, let them identify strengths and weaknesses, increase confidence, and make studying anywhere and anytime easier. For weak-subject focus and consistent study motivation, 46 respondents, or 97.9%, agreed or strongly agreed. These findings should be interpreted as expected or perceived benefits, not as proof that the application improved official Exit Exam pass rates. However, they strongly supported the requirements for flexible practice, feedback, progress awareness, and motivation.

4.1.5 Summary of Findings

The questionnaire results supported the first objective and first research question of this thesis. The respondent profile showed that the sample was directly connected to the Afghan Medical Exit Exam because all respondents were curative medicine or stomatology graduates or final-year students. Although the sample was limited, centered on Kabul, and entirely male, it still provided useful exploratory evidence for requirements elicitation.

The current-preparation results showed that respondents relied on many

preparation sources, especially printed question books or past papers, online groups, university notes or lectures, preparation centers, and self-study. At the same time, on-demand access to practice tests was limited for many respondents, and mobile preparation apps had not been used by most of the sample. These findings supported the need for an accessible preparation system that could bring practice, review, and progress feedback into one mobile workflow.

The internet and offline-access results supported the offline-first design decision. Many respondents had internet access often, but reliability was mixed, and offline access to practice tests was rated as very important or extremely important by most respondents. This means that the core practice experience needed to work without depending on constant connectivity, while online features such as weekly tests, leaderboards, profiles, and challenges could add value when internet access was available.

The feature-interest and perceived-impact results supported the main requirements direction of Exit 360. Offline practice tests, performance analytics, subject-wise weak-area tracking, online weekly tests and leaderboards, and peer-to-peer challenge mode all received positive support. The adoption result was also strong, with every respondent saying that they were likely or very likely to use an Exit Exam preparation mobile app with the proposed features. Overall, the findings justified the requirements defined in Chapter 3 and supported the development direction introduced in Chapter 1.

CHAPTER 5: DISCUSSION

This chapter discusses the meaning of the findings and development outcomes presented in the previous chapters. It connects the questionnaire results, literature review, system requirements, prototype reviews, implementation work, and testing evidence to the main aim of the project. The purpose of the chapter was not to repeat every result from Chapter 4, but to explain what those results suggested about the need for Exit 360 and the value of the solution within its defined scope.

5.1 Main Findings

The findings of this project supported the idea that Exit Exam preparation required a more structured and accessible preparation tool. The problem identified in the earlier chapters was not simply that candidates needed more reading material. The broader issue was that preparation for the Afghan medical Exit Exam involved many subjects, timed performance, repeated practice, review of mistakes, and awareness of progress. The questionnaire results in Chapter 4 therefore mattered because they helped translate this general problem into practical system requirements.

One important finding was the need for more flexible access to practice. The questionnaire results showed that respondents did not always have convenient access to mock tests and regular practice opportunities. This supported the decision to make practice tests available inside the mobile application rather than tying preparation only to physical centers, printed material, or fixed schedules. For a licensing exam that covers a broad subject area, such flexibility was important because candidates needed repeated opportunities to practice, review, and return to weak subjects.

The results also supported the offline-capable direction of the system. Internet access and reliability were important concerns in the Afghan context, and this matched the background evidence discussed in Chapter 1. As a result, the project treated offline preparation as the foundation of the system rather than as an optional extra. The local question bank, SQLite-based attempt storage, offline test creation, local grading, history, and statistics were therefore central to the project. The online layer was designed to add value through accounts, weekly tests, leaderboards, and challenges when connectivity was available, but the basic preparation workflow did not depend on a constant internet connection.

Another finding was that users valued feedback and progress awareness. A preparation system that only showed whether a single test was passed or failed

would not be enough for long-term study. Exit 360 therefore included answer review, attempt history, basic statistics, subject-level performance summaries, and weak-area awareness. These features helped turn repeated practice into a more meaningful study process because users could see where they had performed well and where they needed more revision.

The findings also suggested that online competition could be useful if it remained secondary to the core preparation experience. Weekly tests, leaderboards, and peer challenges were relevant because they could encourage regular practice and give students a sense of comparison. However, the project did not treat competition as a replacement for study. Instead, it was used as an additional motivational layer. This was important because the local context made it unrealistic to build a system that required every candidate to be online at all times.

5.2 Relation to Objectives and Research Questions

The main objective of the project was to design and implement an Afghan medical Exit Exam preparation mobile application. This objective was addressed through the development of Exit 360, which combined a Flutter mobile application, a local SQLite question bank, Firebase-backed online services, and an internal web admin panel. The main research question asked how such an application could be designed and implemented. The project answered this by using a mobile-first and offline-capable architecture, then adding online features through controlled backend services.

The first sub objective was to identify the existing challenges faced by Exit Exam candidates and analyze their main preparation needs. This was addressed through the background study, literature review, review of existing systems, and questionnaire. Together, these sources showed that candidates needed organized subject-wise practice, timed mock tests, answer review, progress tracking, and offline access. The results also showed that online activities could be useful, but only when they complemented the offline preparation workflow.

The second sub objective was to design and implement an Exit Exam preparation mobile application based on the identified requirements. This was addressed by translating the findings into functional, non-functional, and data requirements. The mobile application implemented offline practice, configurable tests, result display, review, attempt history, and statistics. The online side implemented authenticated accounts, weekly tests, leaderboards, friends, and asynchronous challenges. The admin panel supported operational work such as inspecting platform activity, managing weekly-test workflows, reviewing question-bank

summaries, and publishing leaderboards.

The third sub objective was to test and evaluate the developed system in relation to its intended objectives. This was addressed through automated tests, integration testing, emulator-based checks, deployed Firebase testing, and prototype reviews. The prototype reviews were especially important because they connected implementation work with user and developer feedback. The first prototype reviewed the offline preparation flow, the second reviewed the online social and competition layer, and the third reviewed the internal admin workflow. These reviews did not prove long-term educational effectiveness, but they helped evaluate whether the system direction and implemented workflows matched the intended objectives.

5.3 Relation to Literature and Existing Systems

The discussion of Exit 360 was also connected to the literature and existing systems reviewed in Chapter 2. The educational literature supported the value of repeated testing and feedback. As emphasized by Parker (2020), repeated low-stakes testing has learning value, while according to Zhang et al. (2023), realistic practice exams could help learners recognize gaps and prepare more actively. These ideas were reflected in Exit 360 through subject-wise tests, general practice tests, timed sessions, automatic grading, and review of previous attempts.

The mobile learning literature also supported the project direction. According to Rüth et al. (2021), mobile quiz tools could support flexible self-regulated learning, and according to Lazarinis et al. (2017), mobile self-assessment allowed students to practice and receive feedback outside formal classroom settings. This was relevant to Exit 360 because many candidates needed preparation support that could fit into different study times and access conditions. A mobile application was therefore a suitable form for the proposed solution.

The review of local systems showed that Afghan medical exam preparation applications were already emerging, but the landscape was still incomplete. MedQuiz showed local interest in digital medical exam preparation, but its online-first and paid access model limited what users could evaluate freely (Azmoonak Tebi, 2026). EXIT Medical Exam Prep provided a strong offline example with bilingual question access, timed tests, and answer review, but it remained limited in long-term analytics and online competition (Zahid apps store, 2026). Exit 360 therefore did not claim to be the only preparation app. Its contribution was the attempt to combine offline access, progress awareness, Exit-focused practice, and optional online competition in one coherent workflow.

The foreign systems reviewed in Chapter 2 showed a more mature direction for licensing exam preparation. UWorld treated practice as a full cycle of questions, timed blocks, explanations, assessment, and review (UWorld, 2026). AMBOSS connected question practice with medical reference material and performance signals (AMBOSS, 2026). SMLE Gate showed how a focused national licensing-exam app could support simulated exams, notes, saved questions, result review, and progress tracking (SMLE Gate, 2026). These systems helped clarify useful design patterns, but they also showed the difference between large commercial platforms and a student capstone project. Exit 360 adapted the practical ideas that matched the Afghan context while leaving multimedia-heavy content, large-scale coaching, and advanced commercial features outside the project scope.

The literature on online learning conditions in Afghanistan further supported the offline-first approach. In the study by Gawhari et al. (2021), limitations were identified in relation to internet access, connection speed, cost, and digital learning materials. This matched the design decision to keep core preparation local while treating Firebase-backed features as added value. The discussion therefore confirmed that the technical design of Exit 360 was not only a development preference. It was a response to the educational and infrastructural conditions described in the earlier chapters.

5.4 Practical Implications of Exit 360

For Exit Exam candidates, the practical implication of Exit 360 was that preparation could become more organized and repeatable. Instead of depending only on scattered notes, printed questions, or occasional mock tests, students could use the application to select subjects, take timed or untimed tests, review answers, and monitor their progress. This did not replace textbooks, teachers, or preparation centers, but it provided a structured support tool that could be used alongside them.

The offline-first design had a direct practical value. Since the main practice workflow was stored locally, the app could remain useful when the internet was unavailable or unreliable. This was important for a target group that may not always have stable access to online platforms. At the same time, the optional online layer allowed candidates to participate in weekly tests, leaderboards, and challenges when internet access was available. In this way, the system balanced accessibility with motivation.

Technically, the project showed how a student team could combine local mobile storage with cloud-backed trusted workflows. Flutter and SQLite

supported the offline mobile layer, while Firebase Authentication, Firestore, Cloud Functions, and Firebase Cloud Messaging supported account-based and competitive features. Cloud Functions were especially important because they kept trusted operations such as question locking, grading, challenge updates, and leaderboard publishing under backend control rather than relying on the mobile client.

The internal admin panel also had practical importance. It allowed controlled operational work without requiring direct Firestore editing for every task. This mattered because online weekly tests, leaderboard publishing, question-bank summaries, and platform inspection require careful handling. The admin panel therefore supported the maintainability and operational side of the system, even though it remained within the scope of a capstone prototype.

From a Software Engineering perspective, the project demonstrated the value of connecting requirements elicitation, system design, implementation, testing, and documentation. The questionnaire helped identify user needs, the literature review helped shape feature direction, the prototypes created reviewable working stages, and the testing process helped verify important logic and integration points. This made Exit 360 more than a simple mobile app because the software was supported by a documented engineering process.

5.5 Limitations of the Findings and System

The discussion of Exit 360 must also be understood in light of the project's limitations. The questionnaire provided useful requirements-oriented evidence, but it did not represent all Exit Exam candidates in Afghanistan. The achieved sample size was smaller than the statistically recommended sample, and the respondents were selected through a convenience sampling approach. Therefore, the findings were useful for guiding system requirements and early evaluation, but they should not be treated as a complete national measurement of all candidates' needs.

The content limitation was also important. The current question bank was limited in size and was prepared in Dari. A complete bilingual Dari and Pashto question bank would require expert medical review, careful content authorization, and long-term maintenance. The project avoided direct question mining from copyrighted books or unauthorized sources, which was appropriate from an ethical and academic perspective. However, this also meant that the current question bank could not be presented as a full replacement for official or expert-reviewed preparation materials.

The system was also limited by its prototype-level deployment. The project implemented and tested important offline and online workflows, but it was not deployed as a mature production platform with large-scale users, complete monitoring, complete notification coverage, and full app-store release. Firebase deployment also introduced a practical constraint because Cloud Functions required upgrading from the free Spark plan to the Blaze plan, and Google Cloud Billing did not support Afghan payment methods. This showed that technical implementation can be affected by regional service and payment limitations, not only by programming difficulty.

Another limitation was that the project did not measure long-term learning effectiveness or official exam outcomes. The questionnaire, prototype reviews, and tests supported the relevance, usability direction, and functional correctness of the system, but they did not prove that using Exit 360 increased official pass rates. Such a claim would require a longer study with a larger sample, controlled comparison, validated question content, and follow-up after the actual Exit Exam.

Finally, some advanced features remained outside the project scope. These included real-time multiplayer testing, continuously updating live leaderboards, large-scale tournaments, video lessons, medical animations, advanced adaptive learning, and a complete production-grade content management system. These exclusions helped keep the project realistic for the available timeline, team size, and resources, but they also showed where future work could expand the system.

Within these limitations, the questionnaire findings, literature review, existing-system analysis, prototype reviews, implementation work, and testing evidence indicated that Exit 360 addressed the identified preparation need within the project scope. It combined offline question practice, timed tests, answer review, progress tracking, optional online competition, and controlled administrative workflows in a locally relevant system. Exit 360 should therefore be understood as a capstone prototype and documented software engineering project, not as a complete national exam platform or a proven replacement for formal preparation resources. Its contribution was in translating a real preparation problem into a working and technically structured solution. The next chapter presents the overall conclusion of the project and recommendations for future work.

CHAPTER 6: CONCLUSION AND FUTURE WORK

This chapter concludes the thesis by summarizing the work completed, identifying the main contributions of the project, reviewing the achievement of the stated objectives, and presenting future enhancements for Exit 360. It brings together the research and development work described in the previous chapters and outlines how the system can be improved beyond the current capstone scope.

6.1 Summary of Work Done

This project designed and implemented Exit 360, an Afghan medical Exit Exam preparation system intended to support curative medicine and stomatology graduates in their exam preparation. The work began with a review of the Exit Exam context, related literature, local and foreign medical exam preparation systems, and a questionnaire-based requirements elicitation process. These activities helped identify the need for structured practice, offline access, timed testing, answer review, progress awareness, and optional online competition.

Based on these requirements, the project produced a Flutter mobile application with a local SQLite question bank, subject-wise and mixed practice tests, timed sessions, automatic grading, answer review, attempt history, and basic statistics. The system also included a Firebase-backed online layer for user accounts, weekly tests, leaderboard snapshots, friends, challenges, notifications, and trusted backend operations through Cloud Functions. In addition, an internal React, TypeScript, and Vite admin panel was developed to support controlled operational tasks such as question-bank management, weekly-test management, leaderboard publishing, and platform inspection.

The project also produced supporting software engineering artifacts. These included functional, non-functional, and data requirements; use case, sequence, architecture, entity-relationship, and relational schema diagrams; prototype review findings; testing documentation; and a LaTeX thesis that packaged the research and development work. The system was evaluated through questionnaire results, prototype-based user acceptance review, unit testing, integration testing, Firebase Emulator Suite checks, and deployed Firebase function testing.

6.2 Key Contributions

The academic contribution of the project was the documentation of a locally focused software engineering study for Afghan medical Exit Exam preparation. The thesis connected the exam preparation problem with literature review, existing-system analysis, questionnaire-based requirements evidence, system design, implementation, testing, and discussion of limitations. This provided a structured academic record of how a preparation problem was translated into a software solution.

The practical contribution was the development of an offline-capable preparation workflow for Exit Exam candidates. The mobile application allowed students to practice questions, configure tests, use timed sessions, view results, review answers, check attempt history, and observe basic performance statistics. These features addressed the need for repeated and organized preparation without making constant internet access a requirement.

The technical contribution was the integration of local mobile storage with cloud-backed online workflows. Flutter and SQLite supported the offline preparation layer, while Firebase Authentication, Cloud Firestore, Firebase Cloud Functions, and Firebase Cloud Messaging supported account-based features, competitive tests, challenges, notifications, and trusted backend logic. The separate admin panel added operational support for managing online workflows without placing admin-level actions inside the student application.

The process contribution was the use of an iterative development approach supported by prototype reviews and testing. The system was developed through major stages focused on offline preparation, online competition, and admin workflows. Each prototype review helped identify practical improvements, while automated and manual testing helped verify important app logic, backend behavior, and integration points.

6.3 Achievement of Objectives

The objectives defined in Chapter 1 were used as the basis for evaluating whether the project achieved its intended purpose. Table 6.1 summarizes the achievement status of the main objective and sub objectives. In the table, Objective restates each approved project objective, Status indicates whether it was completed within the defined scope, and Evidence identifies the principal research, development, or testing outputs that demonstrated its achievement.

Table 6.1: Objective Achievement Status

Objective	Status	Evidence
To design and implement an Afghan medical Exit Exam preparation mobile application	Completed	Implemented mobile app, Firebase backend, admin panel, requirements, design artifacts, testing, and thesis documentation
To identify the existing challenges faced by Exit Exam candidates and analyze their main preparation needs	Completed	Literature review, existing-system review, questionnaire design, questionnaire analysis, and Chapter 4 results
To design and implement an Exit Exam preparation mobile application based on the identified requirements	Completed	Functional requirements, non-functional requirements, Flutter app, SQLite database, Firebase backend, and React admin panel
To test and evaluate the developed system in relation to its intended objectives	Completed	Unit tests, integration testing, Firebase emulator checks, deployed function testing, and prototype-based user acceptance reviews

As shown in Table 6.1, the project completed its stated objectives within the defined capstone scope. The main objective was achieved through the design and implementation of the Exit 360 mobile application, its Firebase-backed online services, the internal admin panel, and the research, design, testing, and documentation presented in this thesis.

The first sub objective focused on identifying the challenges faced by Exit Exam candidates and analyzing their preparation needs. It was achieved through the literature review, examination of existing local and international systems, questionnaire design and distribution, and analysis of the collected responses presented in Chapter 4.

The second sub objective concerned translating the identified needs into a designed and implemented system. It was achieved through the functional,

non-functional, and data requirements, followed by the implementation of the Flutter mobile application, SQLite offline database, Firebase backend, and React admin panel.

The third sub objective required the developed system to be tested and evaluated in relation to its intended purpose. It was achieved through automated unit tests, integration testing, Firebase Emulator Suite checks, testing against deployed Cloud Functions, and prototype-based user acceptance reviews.

The completed status reported in Table 6.1 should nevertheless be understood within the limitations discussed in Chapter 5, especially the limited sample size, limited question bank, and prototype-level deployment. Completion therefore indicates that the objectives were fulfilled within the approved project boundaries, not that the system had become a complete national platform or had proven long-term effects on official examination results.

6.4 Future Enhancements

The first important future enhancement is the expansion of the question bank. The current question bank is limited and was prepared mainly to support the implemented workflows. A future version should include a larger, expert-reviewed, and properly authorized bilingual Dari and Pashto question bank. Medical teachers and subject specialists should be involved so that questions are accurate, properly categorized, and aligned with the Exit Exam subject structure.

The mobile application can also be improved through stronger polish and user-support features. Future work should improve onboarding, profile completion, authentication messages, password recovery, email verification, notification coverage, and overall interface consistency. The practice section can also be improved with better review filters, improved attempt management, stronger weak-area summaries, and clearer progress visualization.

The online competition layer should be expanded carefully. Weekly tests, leaderboards, friends, and challenges were implemented as core online workflows, but they can be refined further through better navigation, clearer friend and challenge displays, richer leaderboard information, improved notification handling, and stronger user experience around scheduled tests and published results. These improvements should preserve the offline-first foundation and avoid making the app dependent on constant connectivity.

The admin panel also requires future development. Important improvements

include direct question upload and editing, subject-selection dropdowns instead of manual internal identifiers, better admin account management, stronger filtering, audit logs, and safer operational tools. These features would reduce the need for direct Firestore manipulation and make the platform easier to operate if it is used beyond the prototype stage.

Future research and evaluation should involve a larger and more diverse respondent group. A broader study could include more Exit Exam candidates from different provinces, universities, and preparation backgrounds. A longer-term evaluation could also study how students use the app over time and whether regular use is associated with improved preparation confidence, practice frequency, or official exam performance. Such claims were outside the scope of the current project, but they are important for future validation.

Finally, future work should prepare the system for production deployment. This would include app-store release preparation, monitoring, backup planning, cost planning for Firebase services, security review, content governance, institutional collaboration, and a clearer maintenance plan. These steps would be necessary before Exit 360 could move from a capstone prototype into a larger public preparation platform.

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APPENDIX A: QUESTIONNAIRES

A.1 Requirements Elicitation Questionnaire

This section presents the Persian questionnaire used during the requirements elicitation phase of the project. The questionnaire was prepared for medical graduates and final-year students preparing for the Afghan Medical Exit Exam, since they represent the intended users of Exit 360. It was distributed in Persian to make the questions clear and accessible for respondents. The collected responses were used to understand candidates' preparation habits, access to practice tests, internet availability, interest in offline study features, and expected usefulness of the proposed mobile application.

پرسشنامهٔ سروی اگزیت ۳۶۰

ما محصلان سال آخر دیپارتمنت انجنیری نرم‌افزار پوهنتون کابل هستیم و در حال توسعهٔ یک برنامهٔ موبایل برای آمادگی امتحان اگزیت، به نام اگزیت ۳۶۰، منحیث پروژهٔ نهایی خود می‌باشیم. این سروی محرمانه دربارهٔ دسترسی محصلان و فارغان طب به امتحانات تمرینی، عادات مطالعه و علاقه‌مندی شان به یک اپلیکیشن تمرینی آفلاین و رقابتی می‌پرسد. اشتراک اختیاری است و هیچ‌گونه معلومات شخصی قابل شناسایی نشر نخواهد شد.

بخش ۱: مشخصات فردی

۱. سن شما چند است؟ _____

۲. جنسیت شما چیست؟
الف) مذکر ب) مؤنث

۳. شما اصلاً از کدام ولایت هستید؟ _____

۴. سطح تحصیلی شما چیست؟
الف) فارغ طب معالجوی ب) محصل سال آخر طب معالجوی
ج) فارغ ستوماتولوژی د) محصل سال آخر ستوماتولوژی ه) دیگر _____

۵. آیا قبلاً امتحان اگزیت طبی افغانستان را سپری نموده‌اید؟
الف) بلی، یک بار ب) بلی، بیشتر از یک بار ج) نخیر، هنوز نه

۶. آیا فعلاً در کدام کورس یا برنامهٔ آمادگی امتحان اگزیت درس می‌خوانید؟
الف) بلی ب) نخیر

بخش ۲: تمرین و دسترسی فعلی

۷. شما فعلاً برای امتحان اگزیت چگونه آمادگی می‌گیرید؟ (همه گزینه‌های مناسب را انتخاب کنید)
الف) مطالعهٔ شخصی ب) مرکز آمادگی ج) کورس آمادگی آنلاین د) برنامهٔ موبایل آمادگی
ه) کتاب‌های سؤالات یا سؤالات سال‌های گذشته و) نوت‌ها یا لکچرهای پوهنتون
ز) گروپ‌های تلگرام، واتساپ یا دیگر گروپ‌های آنلاین ح) دیگر _____

۸. هر چند وقت یک بار امتحان آزمایشی یا تمرینی اگزیت می‌دهید؟
الف) هرگز ب) یک بار در ماه ب) چند بار در ماه ج) یک بار در هفته ب) چند بار در هفته د) هر روز

۹. دسترسی شما به امتحان‌های آزمایشی یا تمرینی اگزیت تا چه اندازه آسان است؟
الف) بسیار آسان ب) آسان ج) متوسط د) دشوار ه) بسیار دشوار

۱۰. تا چه حدی برای مطالعه، به انترنت دسترسی دارید؟
الف) هیچ ب) ندرتاً ج) گاهی د) اغلب ه) همیشه

۱

۱۱. کیفیت اتصال انترنت تان را چگونه ارزیابی می‌کنید؟
الف) بسیار بد ب) بد ج) متوسط د) عالی ه) بسیار عالی

۱۲. اهمیت دسترسی آفلاین به امتحان‌های تمرینی را چگونه ارزیابی می‌کنید؟
الف) اصلاً مهم نیست ب) کمی مهم است ج) تا حدی مهم است د) زیاد مهم است ه) بسیار زیاد مهم است

۱۳. آیا تا حال از کدام برنامهٔ آمادگی امتحان اگزیت یا امتحان طبی استفاده کرده‌اید؟
الف) بلی ب) نخیر

۱۴. اگر بلی، از کدام برنامه یا برنامه‌های آمادگی امتحان اگزیت یا امتحان طبی استفاده کرده‌اید؟ (همه گزینه‌های مناسب را انتخاب کنید)
الف) آزمونک طبی (MedQuiz) ب) EXIT Medical Exam Prep ج) دیگر _____

بخش ۳: اولویت سنجی قابلیت‌ها

۱۵. در برنامهٔ اگزیت ۳۶۰، قصد داریم قابلیت‌های ذیل را اضافه نماییم که هر یک مختصراً توضیح داده شده است. بنابراین، این قابلیت‌ها تا چه اندازه برای شما مفید خواهند بود؟ لطفاً به هر قابلیت از ۱ تا ۵ نمره دهید، که در آن ۱ = اصلاً مفید نیست، ۲ = کمی مفید است، ۳ = تا حدی مفید است، ۴ = زیاد مفید است و ۵ = بسیار زیاد مفید است.

قابلیت‌ها	۱	۲	۳	۴	۵
الف) امتحانات تمرینی آفلاین: آزمون‌های کامل را بدون انترنت تمرین کنید					
ب) داشبورد تحلیل عملکرد: آمار دقیق نمرات و پیشرفت خود را در طول زمان بررسی کنید					
ج) شناسایی ضعف‌ها به اساس مضمون: مضمون‌هایی را که به تمرین بیشتر نیاز دارند، تشخیص دهید					
د) پروفایل‌های کاربری و دنبال کردن: حساب بسازید و دوستان تان را پیدا و دنبال کنید					
ه) امتحانات رقابتی آنلاین و جدول رده‌بندی: با دیگران هفته‌وار رقابت کنید و مقام خود را ببینید					
و) چالش میان دوستان: دوستان را برای رقابت با خودتان به امتحانات چالشی دعوت کنید					

۱۶. از فهرست بالا، لطفاً ۳ قابلیت برتر به نظر تان را با استفاده از کودهای حروفی هر قابلیت انتخاب و رتبه‌بندی کنید.

- ✓ رتبهٔ ۱: _____
- ✓ رتبهٔ ۲: _____
- ✓ رتبهٔ ۳: _____

۱۷. اگر یک برنامهٔ موبایل آمادگی امتحان اگزیت با قابلیت‌های فوق در دسترس باشد، چقدر احتمال دارد که از آن استفاده کنید؟
الف) بسیار بعید ب) بعید ج) خنثی د) محتمل ه) بسیار محتمل

۲

بخش ۴: تأثیر محسوس

برای هر جملهٔ زیر، لطفاً سطح موافقت خود را با استفاده از این مقیاس مشخص سازید:
۱ = کاملاً مخالفم، ۲ = مخالفم، ۳ = خنثی، ۴ = موافقم، ۵ = کاملاً موافقم.

شماره	جملات	۱	۲	۳	۴	۵
۱۸	استفاده از این برنامه آمادگی من برای امتحان اگزیت را بهبود می‌بخشد.					
۱۹	استفاده از این برنامه به من کمک می‌کند که بیشتر تمرین کنم.					
۲۰	استفاده از این برنامه به من کمک می‌کند تا نقاط قوت و ضعف خود را تشخیص بدهم.					
۲۱	استفاده از این برنامه به من کمک می‌کند تا روی مضامینی تمرکز کنم که در آن‌ها به تمرین بیشتر نیاز دارم.					
۲۲	استفاده از این برنامه من را تشویق می‌کند که به‌طور منظم‌تر مطالعه کنم.					
۲۳	استفاده از این برنامه اعتماد به نفس من را در شرکت در امتحان افزایش می‌دهد.					
۲۴	استفاده از این برنامه مطالعه را در هر زمان و هر مکان برایم آسان‌تر می‌سازد.					

بخش ۵: تجربه و بازخورد آزاد

۲۵. بزرگ‌ترین موانعی که در آمادگی برای امتحان اگزیت با آن روبه‌رو هستید چیست؟

۲۶. اگر از کدام برنامهٔ آمادگی امتحان اگزیت یا امتحان طبی استفاده کرده‌اید، چه چیزهایی در آن به نظر تان خوب بود و چه چیزهایی نبود؟

۲۷. چه قابلیت‌هایی باعث می‌شود شما یک برنامهٔ آمادگی امتحان را به‌جای روش کاغذی یا روش مبتنی بر کورس انتخاب کنید؟

۲۸. هرگونه نظر یا پیشنهاد دیگری دربارهٔ یک سیستم آمادگی امتحان؟

۳

APPENDIX B: SOURCE CODE REPOSITORY

GitHub Repository: https://github.com/[username]/[project-name]

QR Code: [Insert QR code image]

Installation Instructions:

```
git clone [repository-url]
cd project-name
npm install
npm start
```

APPENDIX C: RAW TEST DATA

Detailed test logs and results.

APPENDIX D: USER MANUAL

5-10 page user guide.